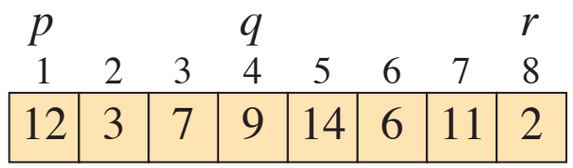
Algorithm Analysis and Elementary Sorting Algorithms

Lecture II for course in Data Structures & Algorithms for AI

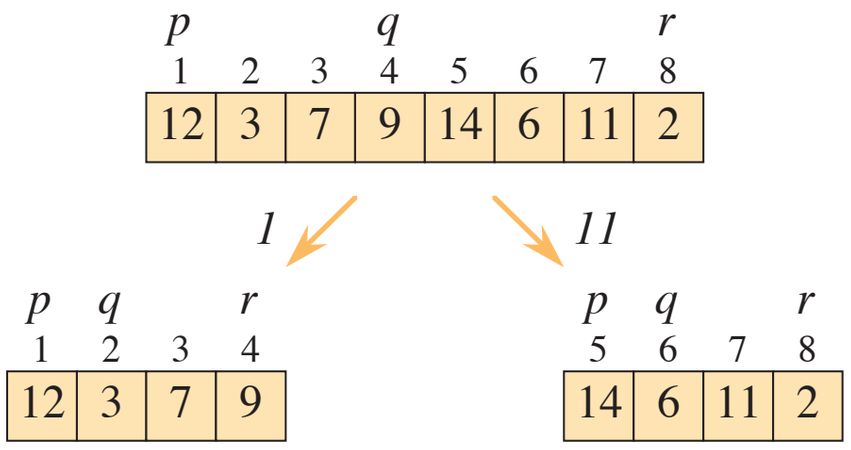
Lectures by Hannah Santa Cruz Baur. Based on *Introduction to Algorithms*, Cormen et al.

Merge sort



Merge sort

divide



- splits an array into two subarrays down the middle,

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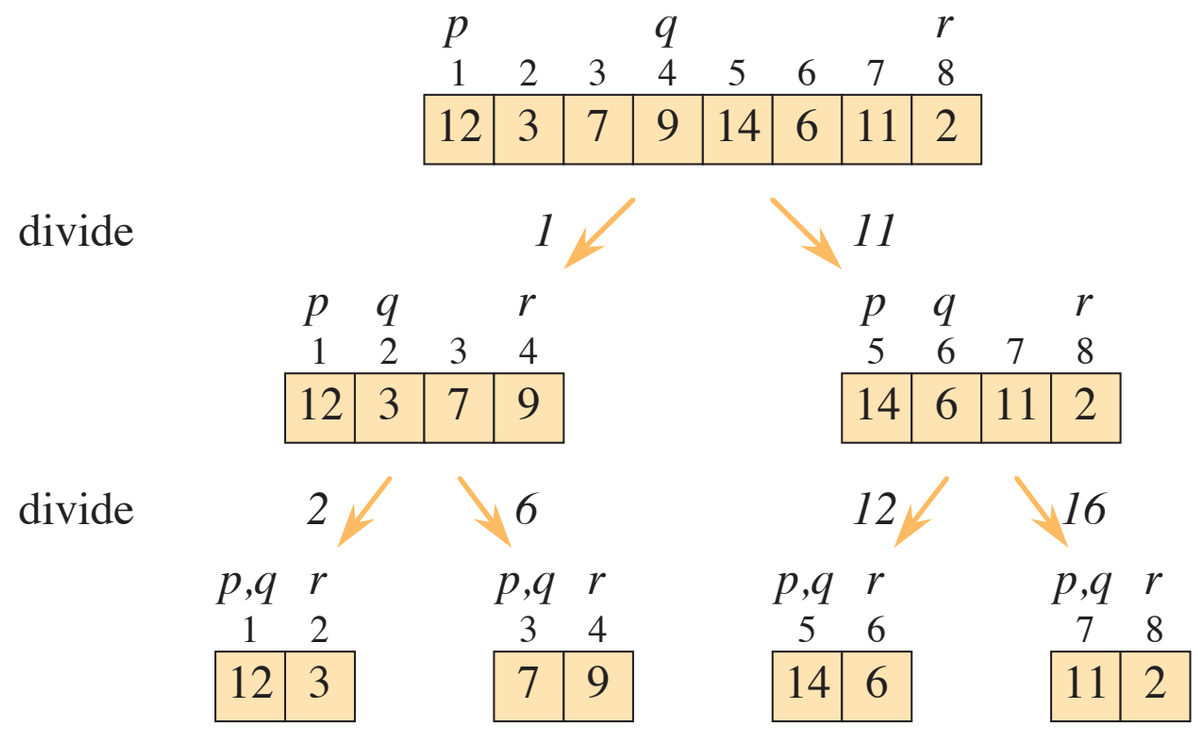

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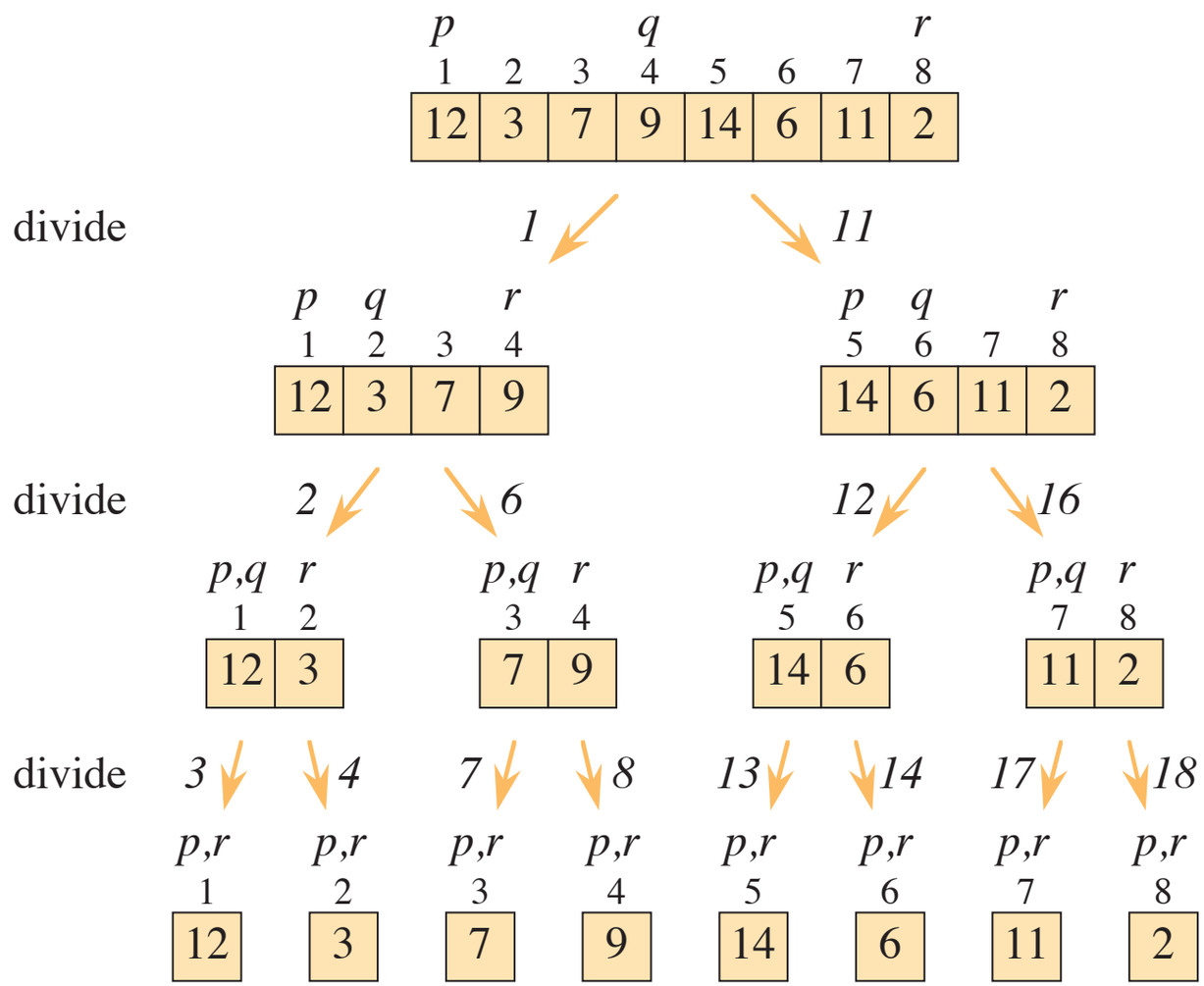

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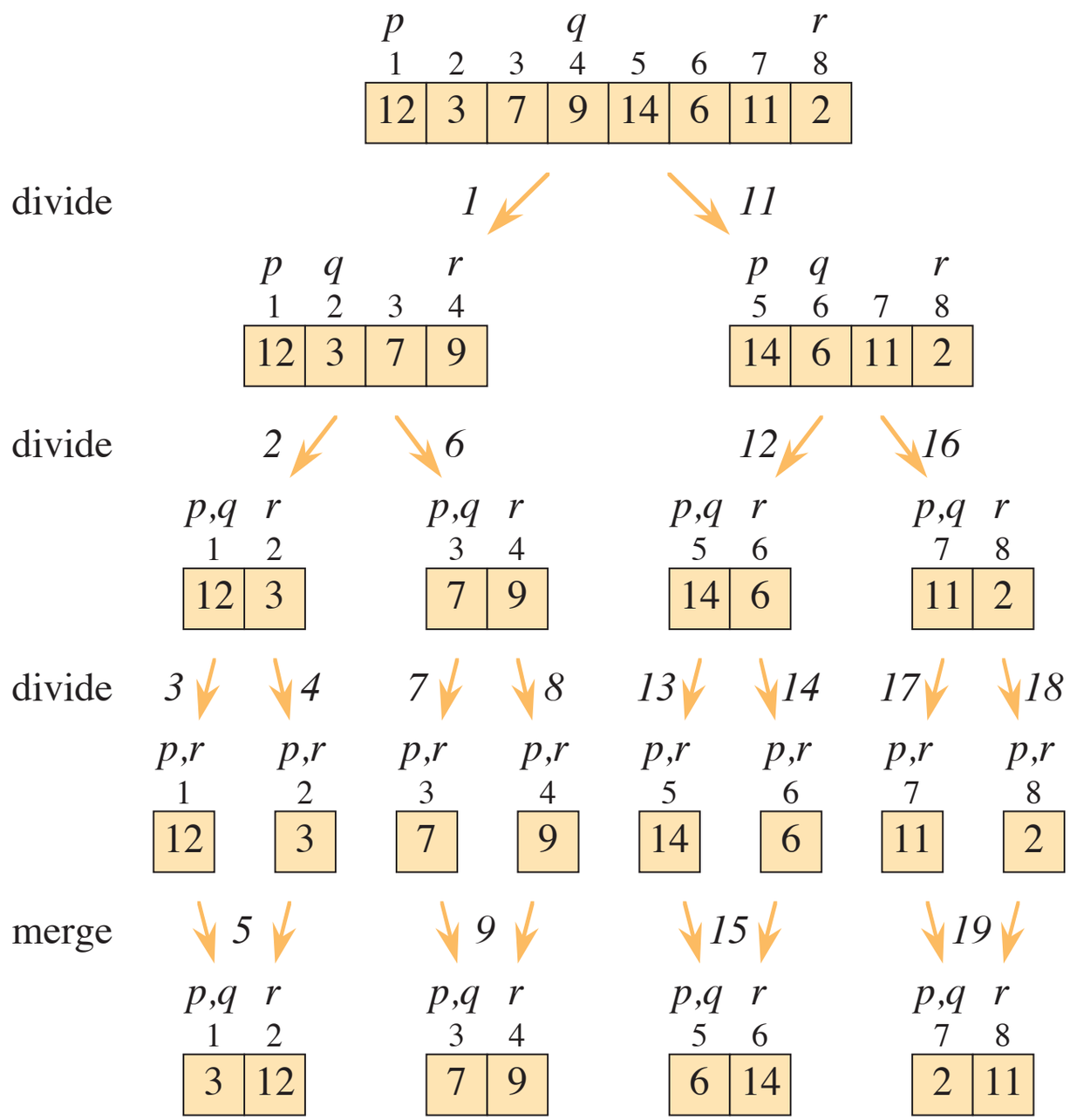

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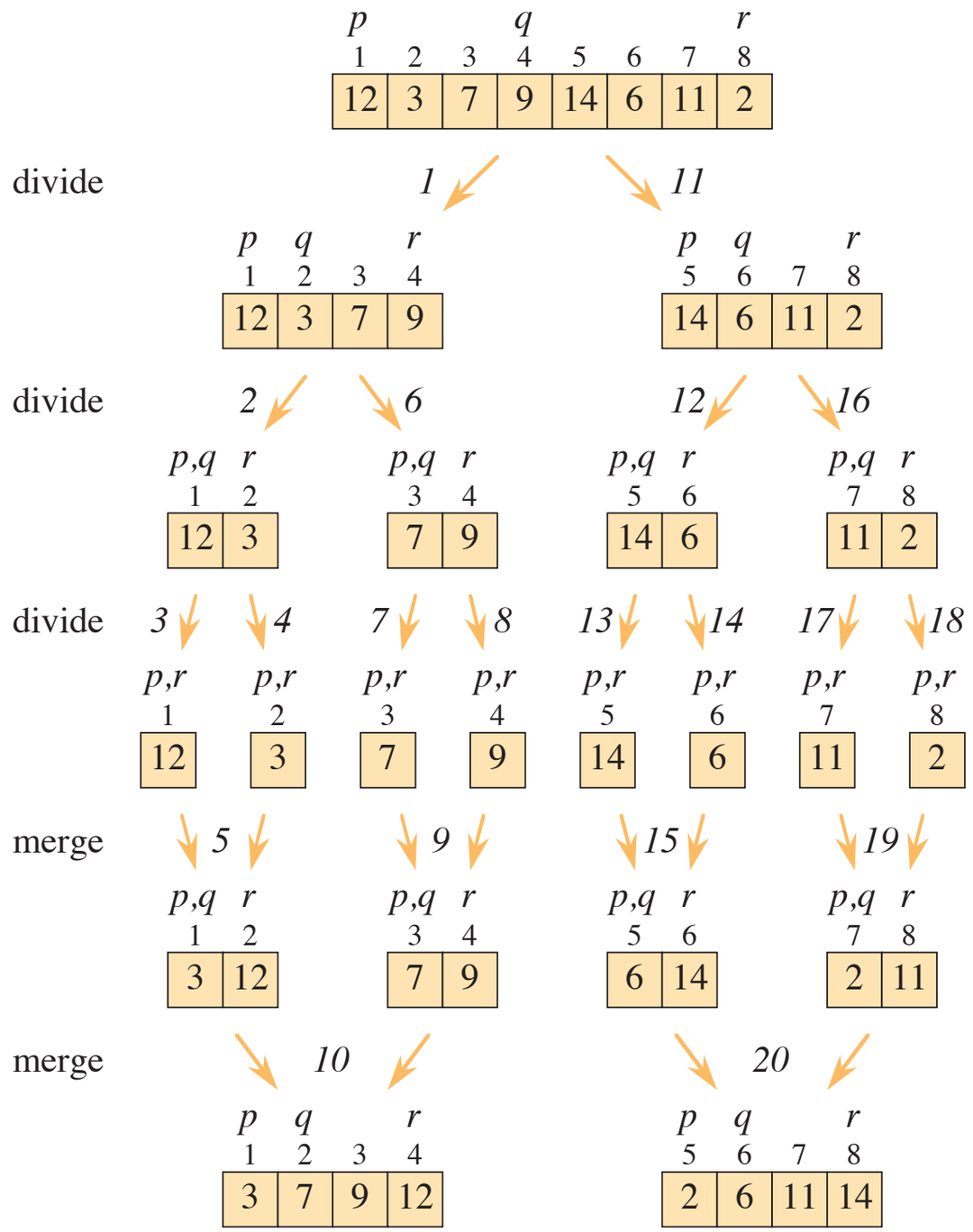

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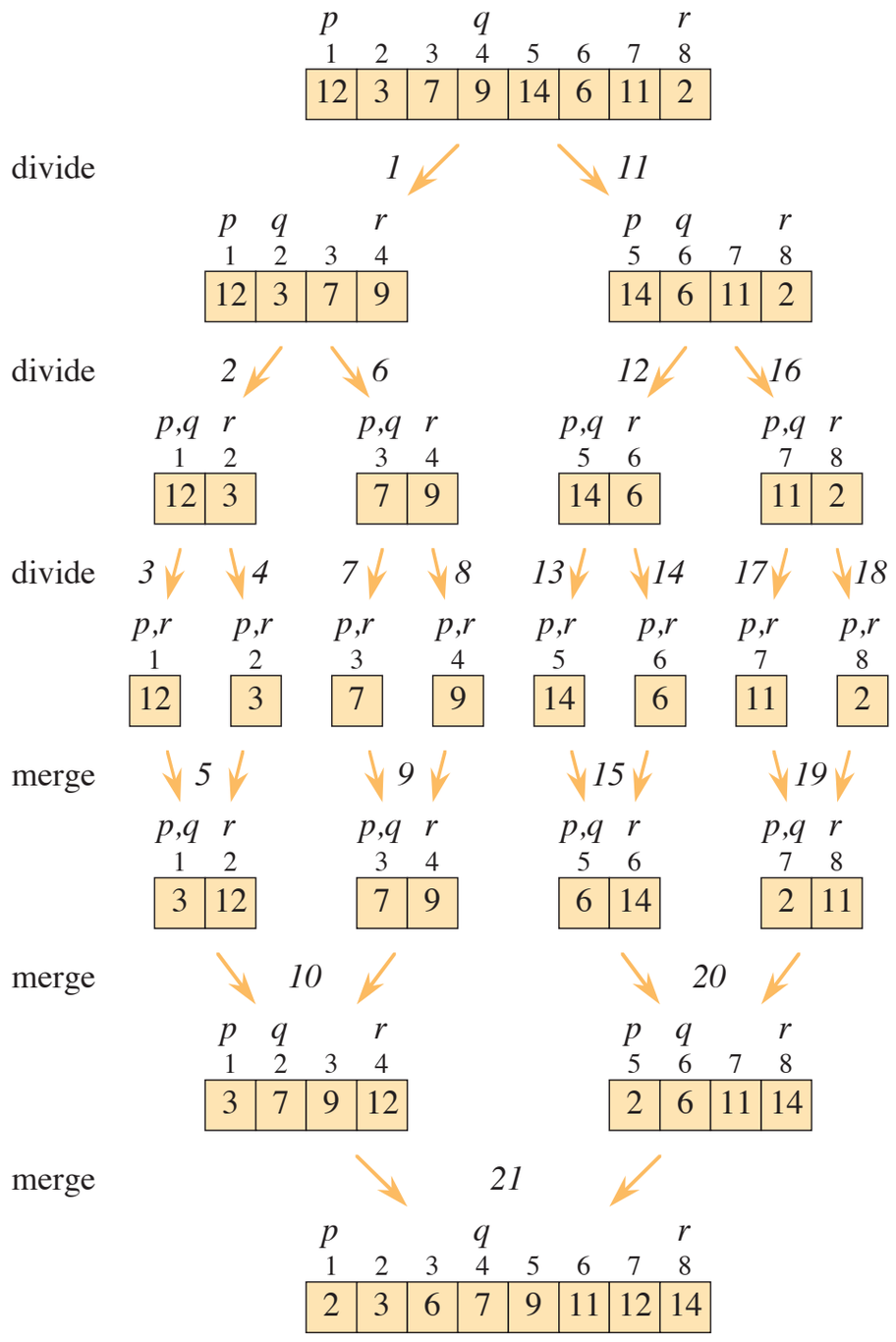


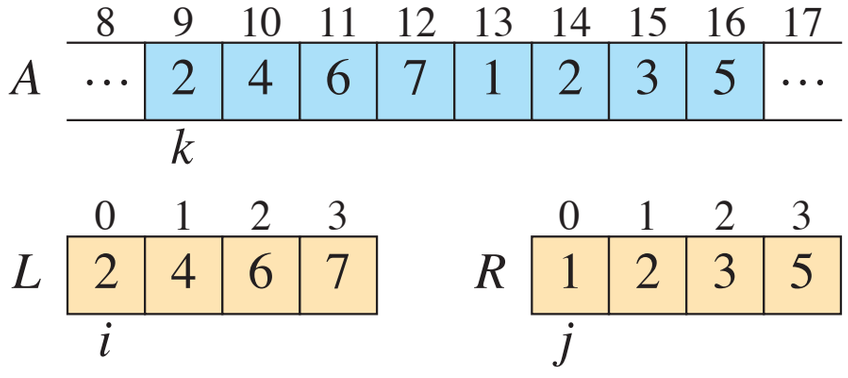
- splits an array into two subarrays down the middle,
- recurses down to smaller and smaller arrays,
- and finally combines the adjacent sorted subarrays, into one singular sorted array.

Merge sort

- Consider an array A, with entries at indexes $p, \ldots, q$ to be sorted

```
MERGE-SORT(A, p, r)
1   if p ≥ r                                  // zero or one element?
2       return
3   q = ⌊(p + r)/2⌋                           // midpoint of A[p : r]
4   MERGE-SORT(A, p, q)                       // recursively sort A[p : q]
5   MERGE-SORT(A, q + 1, r)                   // recursively sort A[q + 1 : r]
6   // Merge A[p : q] and A[q + 1 : r] into A[p : r].
7   MERGE(A, p, q, r)
```

The MERGE procedure



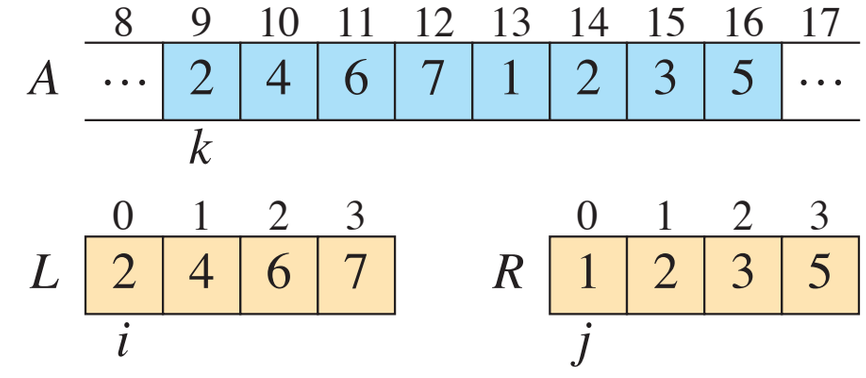
Consider the concatenated array A of two sorted arrays L and R.

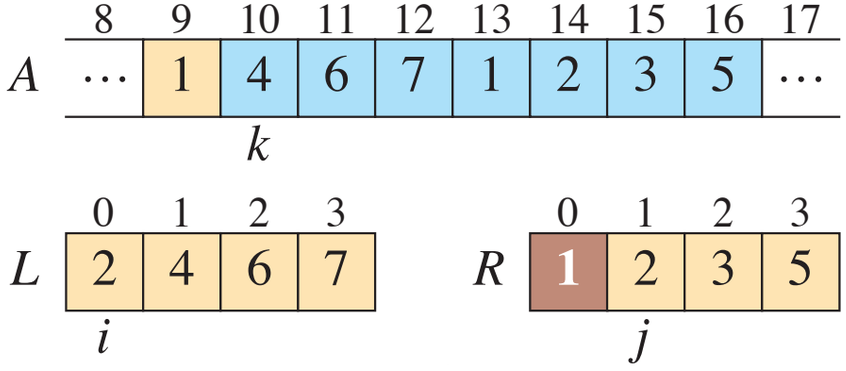
We wish to fully sort the array, by utilizing the fact that L & R are already sorted.

To save storage, we will overwrite the values in A.

We indicate with blue the elements of A that have yet to be overwritten.

The MERGE procedure





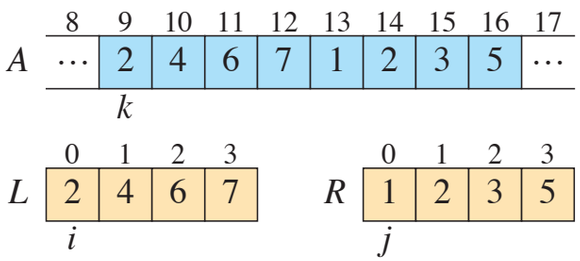
We overwrite the first element of A with the smallest element of L & R.

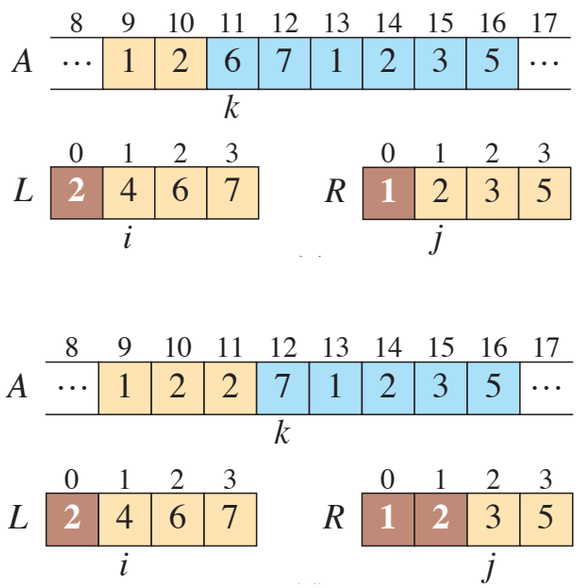
Because L & R are sorted, it suffices to compare their first elements to find the smallest.

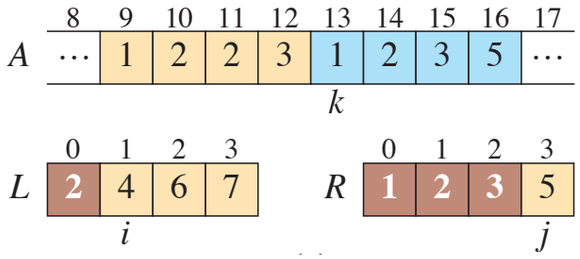
The first element of A is now tan, to indicate it is the final element in its position.

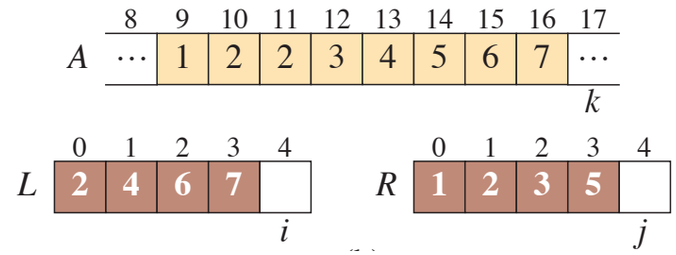
The first element of R is now brown, to indicate it has been copied over.

The MERGE procedure









We continue so, until all elements of A have been overwritten.

This outputs a sorted array A, corresponding to the merger of sorted arrays L & R.

Analysis of merge sort

- We compute the worst case running time of merge sort, $T(n)$, recursively, distinguishing the three stages of the algorithm:
  - **Divide** computing the middle of the array takes constant time $\Theta(1)$.
  - **Conquer** recursively solving two subproblems, each of size $n/2$, takes a total running time of $2T(n/2)$.
  - **Combine** The merge procedure takes $\Theta(n)$ time.
- Adding them together, we get

$$T(n) = 2T(n/2) + \Theta(n) = \Theta(nlog(n))$$

Quick Sort
and
Bucket Sort

Lecture IV for course in
Data Structures & Algorithms for AI

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Divide and conquer algorithms

- Another popular divide and conquer algorithm is Quick sort:
  - **Divide:** partition into two subarrays- such that all elements in one subarray are smaller than a chosen **pivot**, and those in the other subarray are in larger than this pivot.

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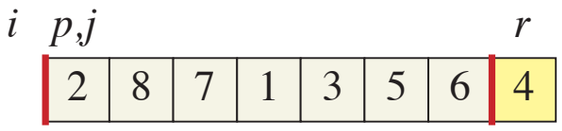
  - **Conquer:** recursively calls itself to sort the smaller arrays.
  - **Combine:** join the two arrays, the union is immediately sorted, there is no need for any more steps.
- Compare with **merge sort**, which:
  - splits an array into two subarrays down the middle,
  - recursively calls itself to sort the smaller arrays,
  - combines the sorted subarrays, by using the MERGE procedure to create one sorted array.

QUICK SORT

- Consider an array A, with entries at indexes $p, \ldots, r$ to be sorted

```
QUICKSORT(A, p, r)
1   if p < r
2       // Partition the subarray around the pivot, which ends up in A[q].
3       q = PARTITION(A, p, r)
4       QUICKSORT(A, p, q − 1)  // recursively sort the low side
5       QUICKSORT(A, q + 1, r)  // recursively sort the high side
```

The PARTITION procedure



The entry at index r will be the pivot.

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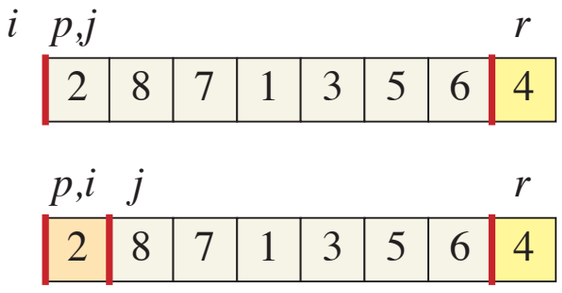

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Tan elements have been added to the "low subarray" - those entries with value smaller than the pivot.

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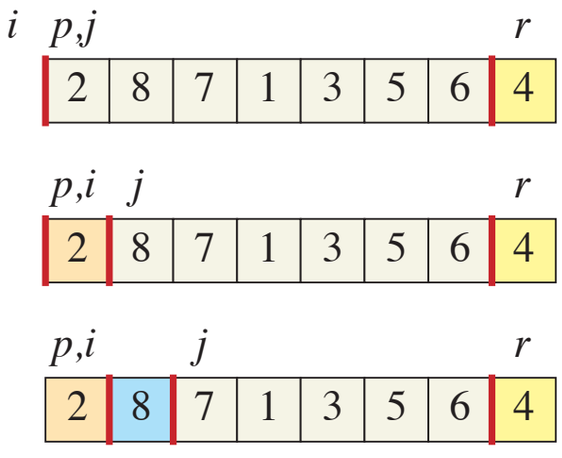

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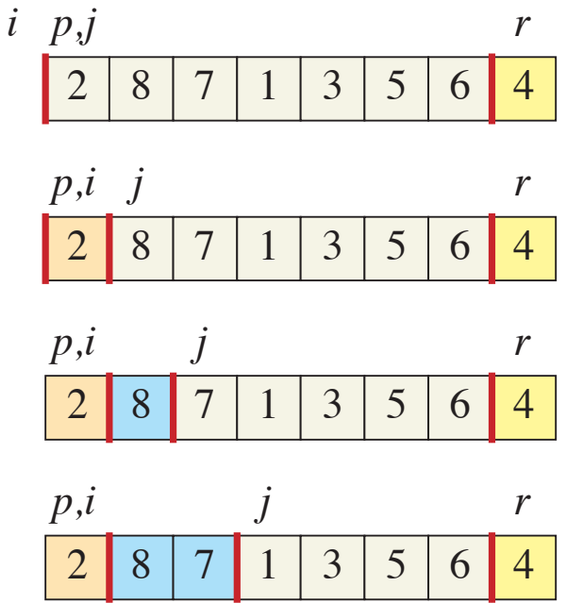

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Elements are combed from left to right, white elements are yet to be classified.

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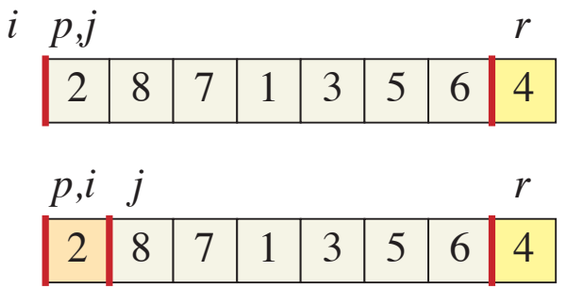

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Value *2* is swapped with itself, being placed in the *low subarray*.

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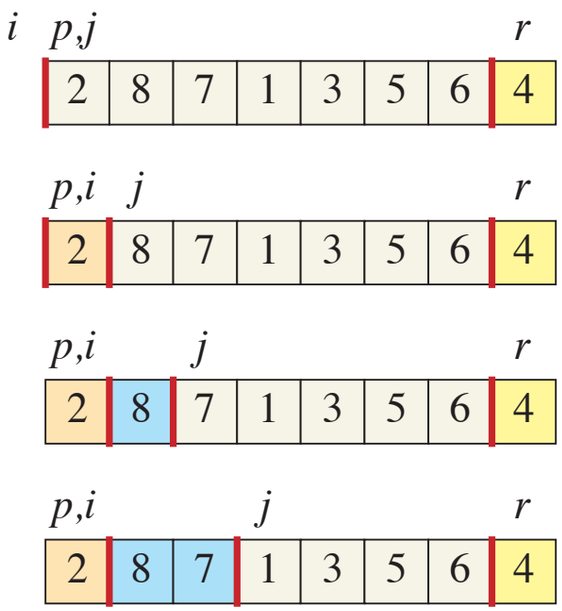

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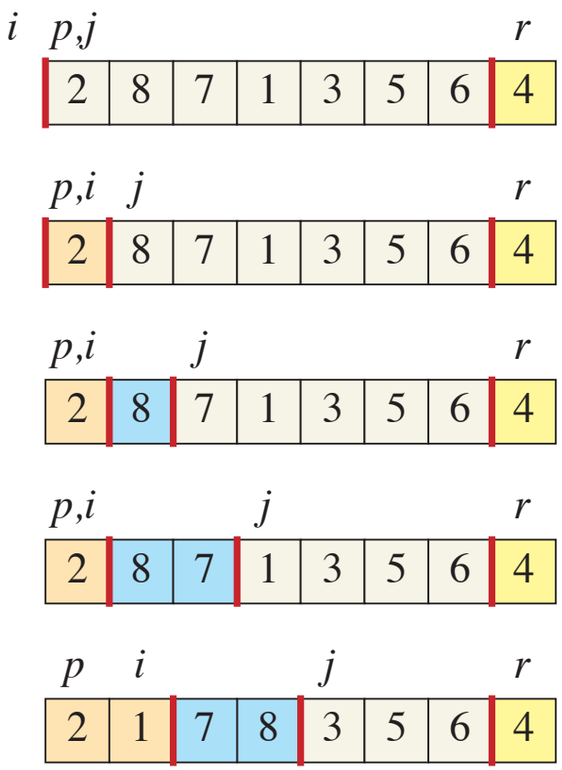

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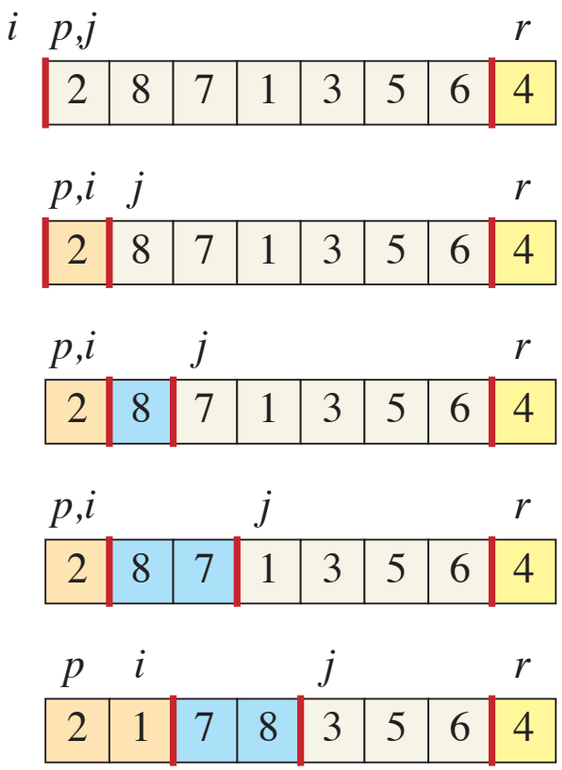

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Notice the position within their subarrays is irrelevant.

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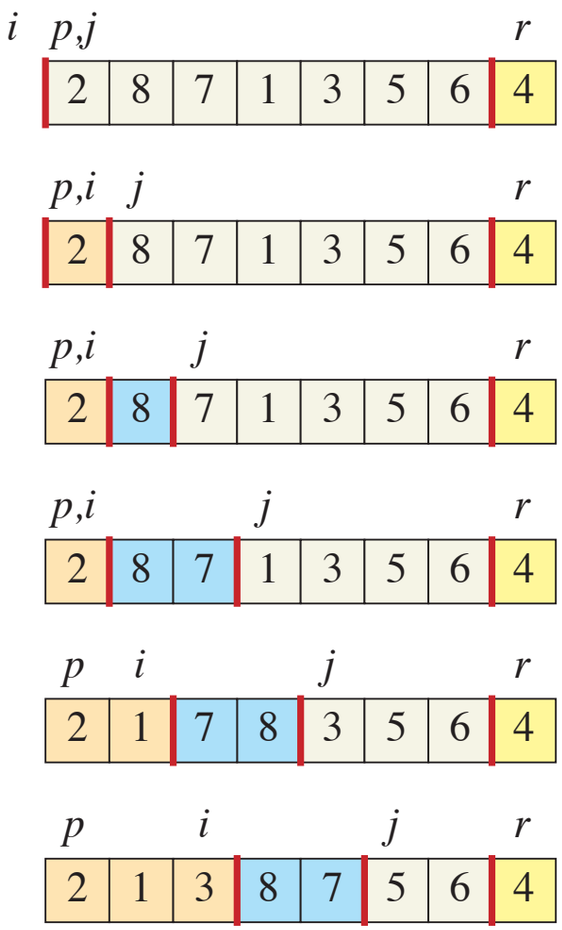

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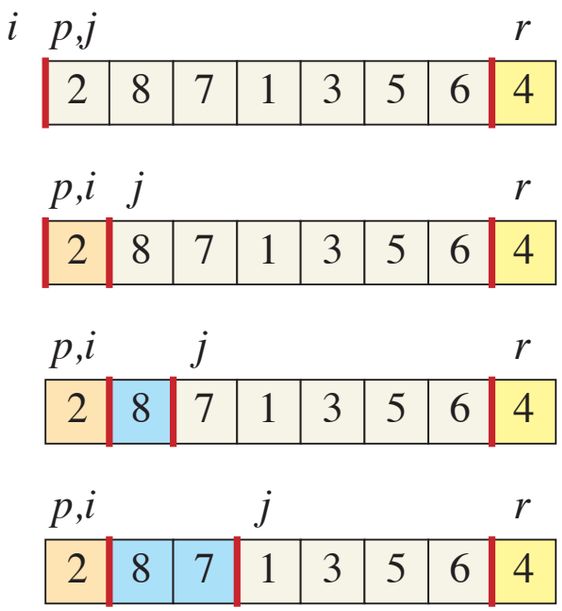

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Continue such ...

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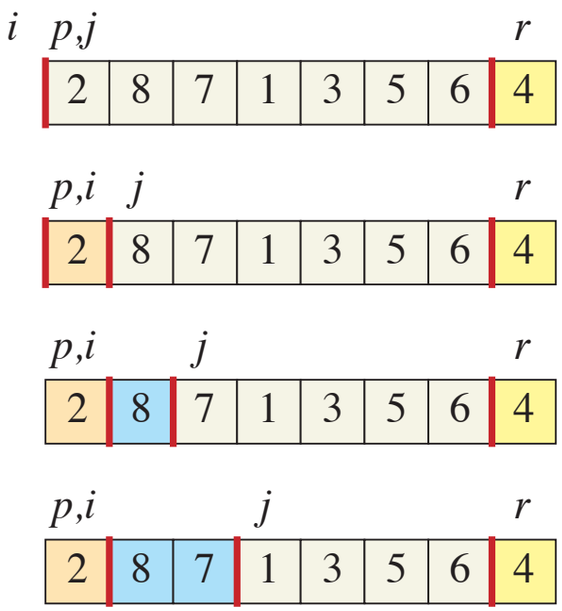

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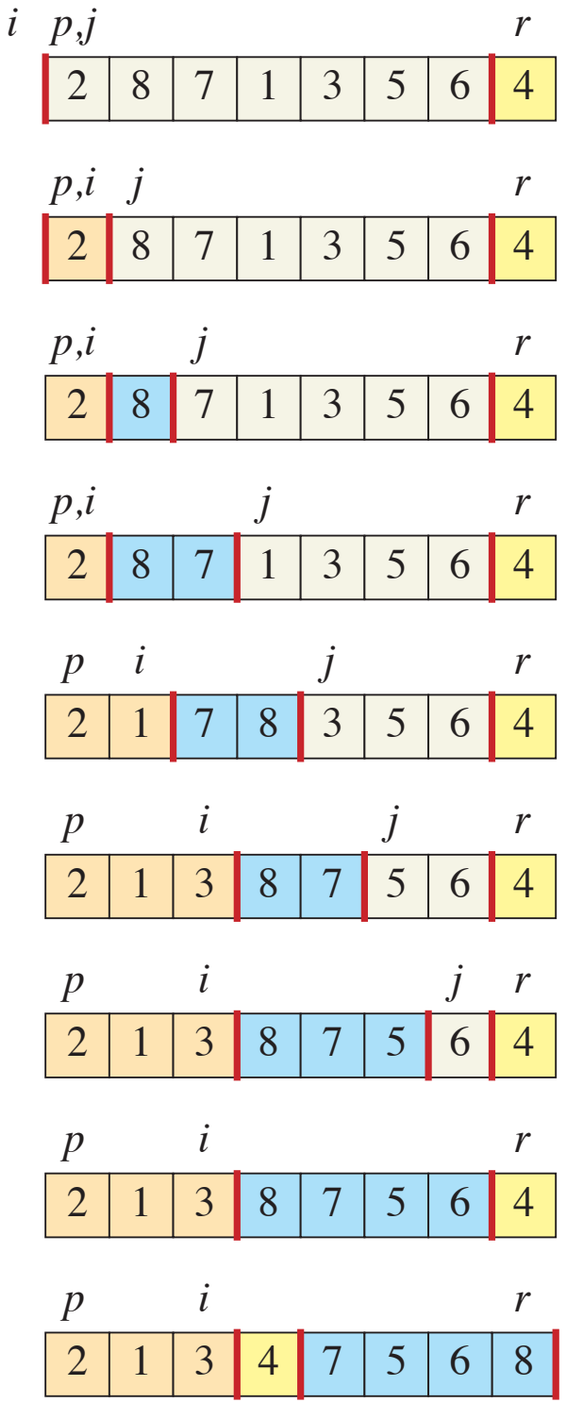
Continue such ... Till the pivot is moved.

The PARTITION procedure



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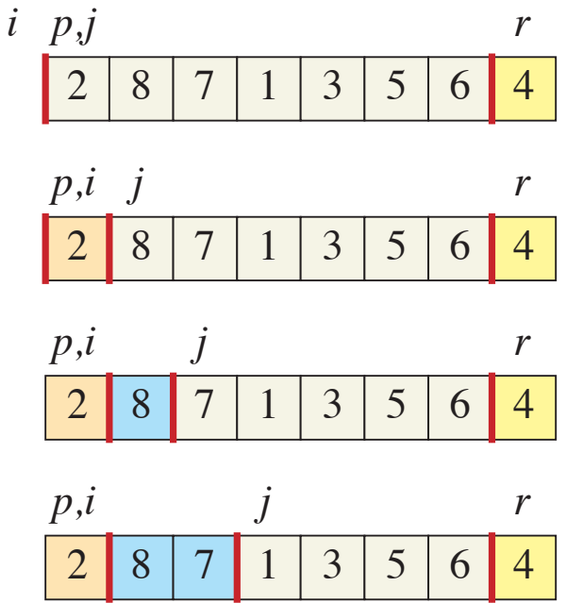
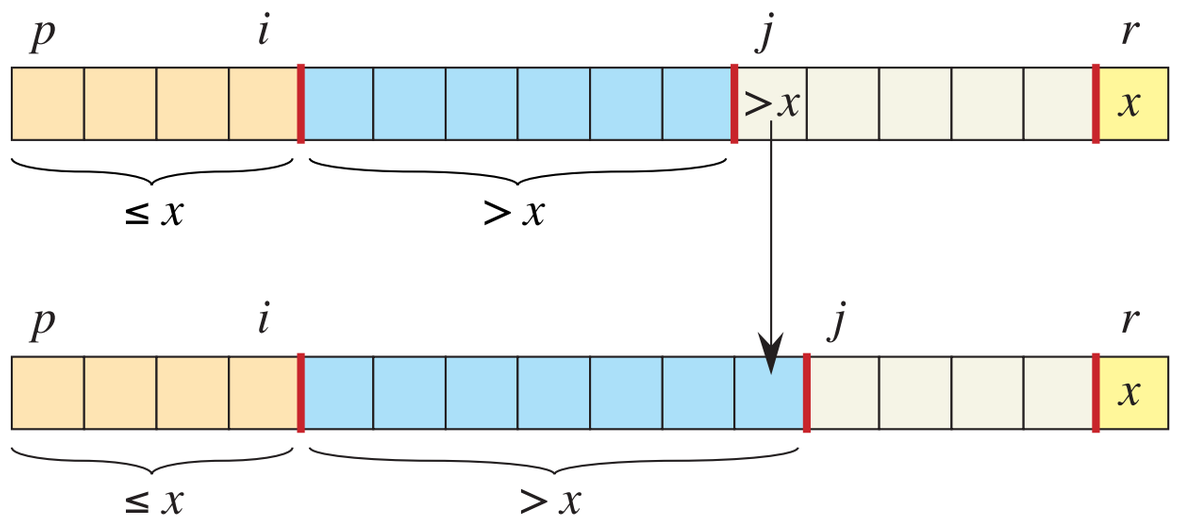


The partition procedure, decides between two possible moves:

- When a value is swapped with itself, to grow the *high subarray*



The only required action is increasing index j.

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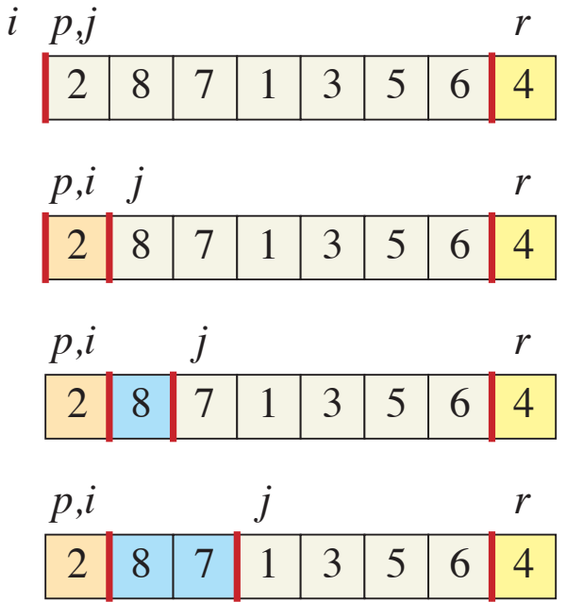

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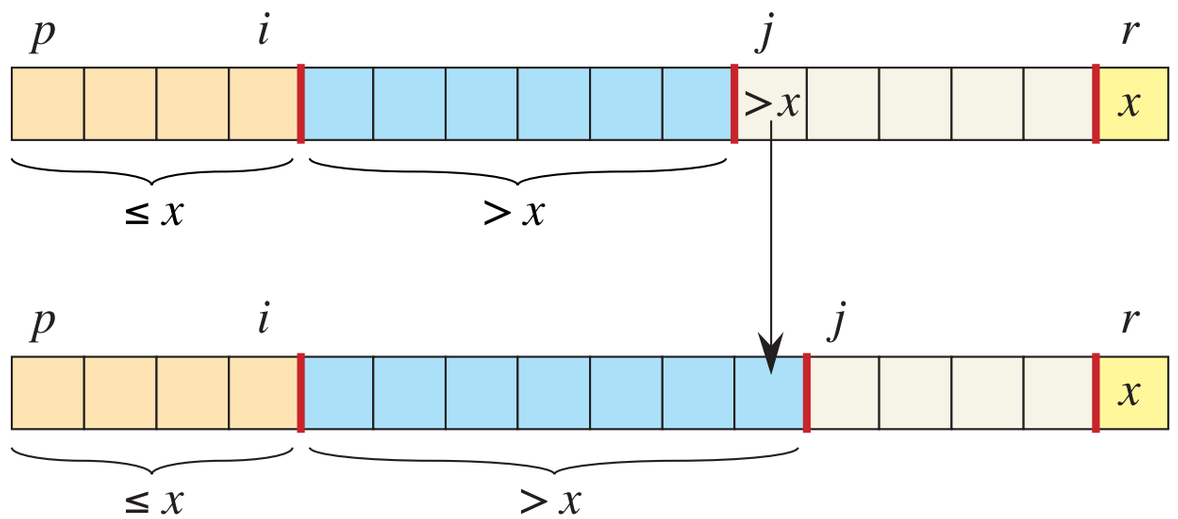

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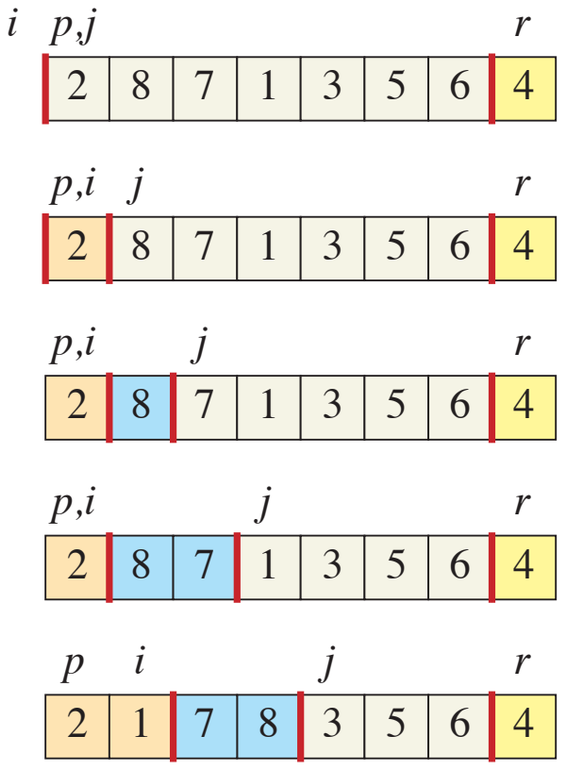
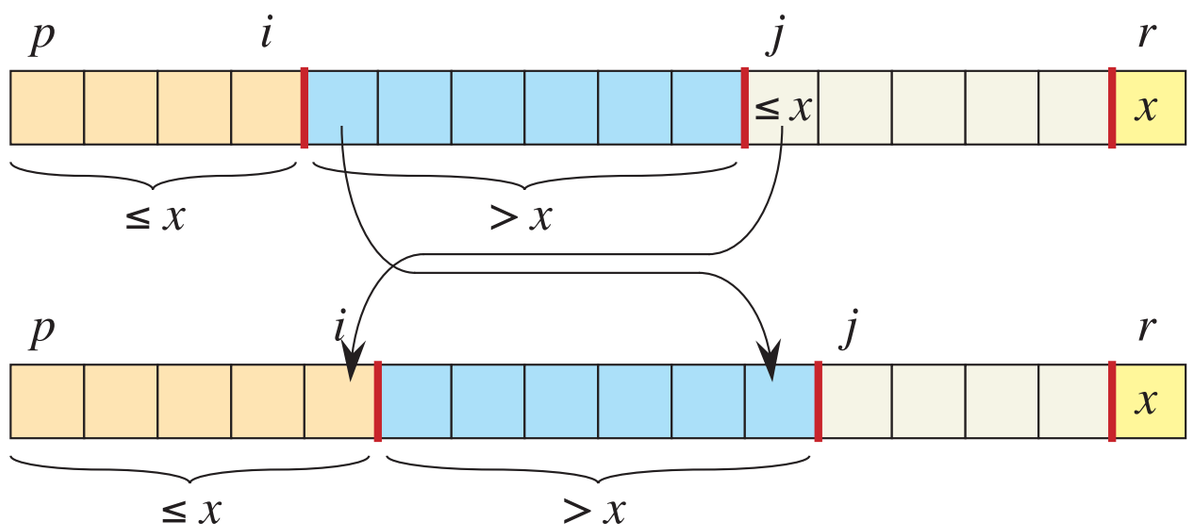


The partition procedure, decides between two possible moves:

- When a value is swapped with itself, to grow the *high subarray* *
- When a value is swapped with another, to grow the *low array*.



Indices i & j are incremented, and the keys are swapped.

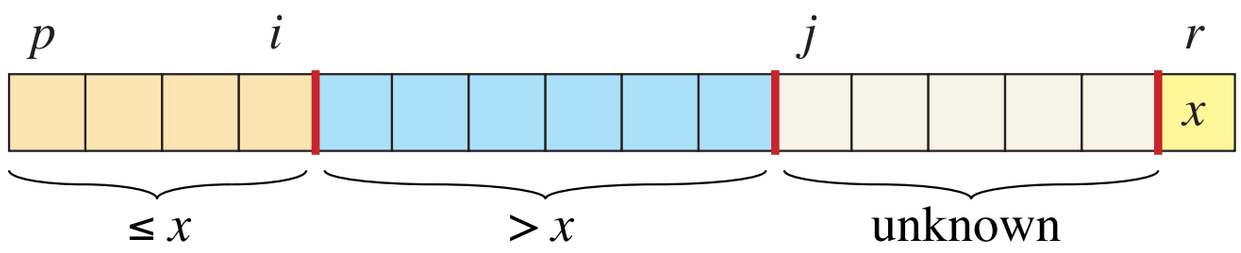
The PARTITION procedure

- Consider an array A, with entries at indexes $p, \ldots, r$ to be sorted.
- We wish to partition the array in place into two subarrays around a pivot (which will be moved, but we will track it).

```
PARTITION(A, p, r)
1  x = A[r]                          // the pivot
2  i = p − 1                         // highest index into the low side
3  for j = p to r − 1                // process each element other than the pivot
4      if A[j] ≤ x                   // does this element belong on the low side?
5          i = i + 1                     // index of a new slot in the low side
6          exchange A[i] with A[j]       // put this element there
7  exchange A[i + 1] with A[r]       // pivot goes just to the right of the low side
8  return i + 1                      // new index of the pivot
```

PARTITION loop invariant

- The partition procedure admits the following loop invariant:
  - If $p \le k \le i$ then $A[k] \le x$
  - If $i + 1 \le k \le j - 1$ then $A[k] > x$
  - If $k = r$ then $A[k] = x$



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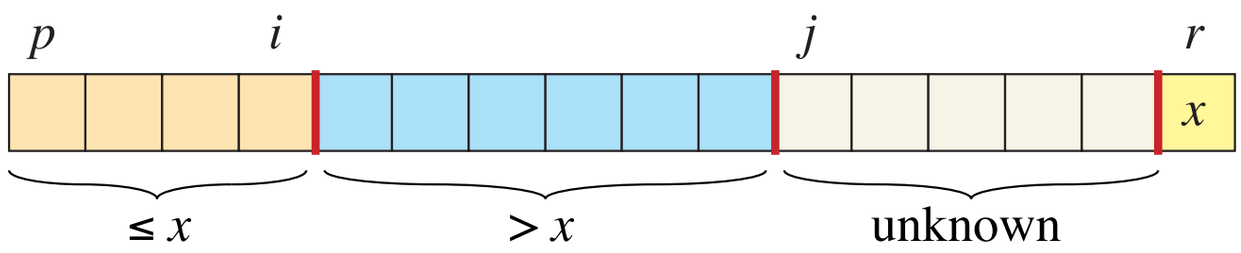

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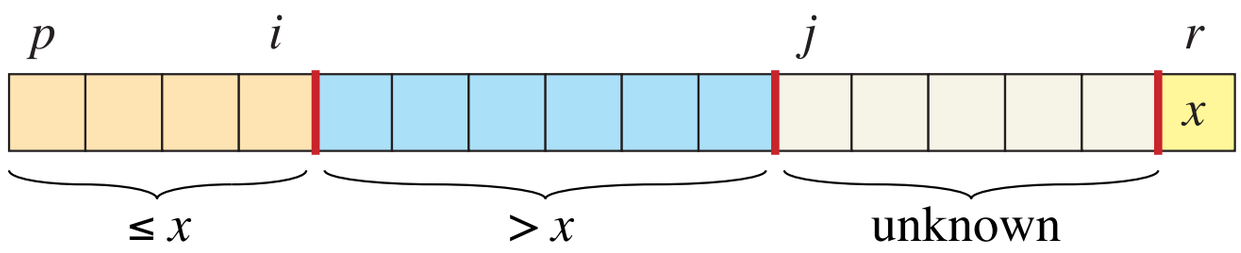

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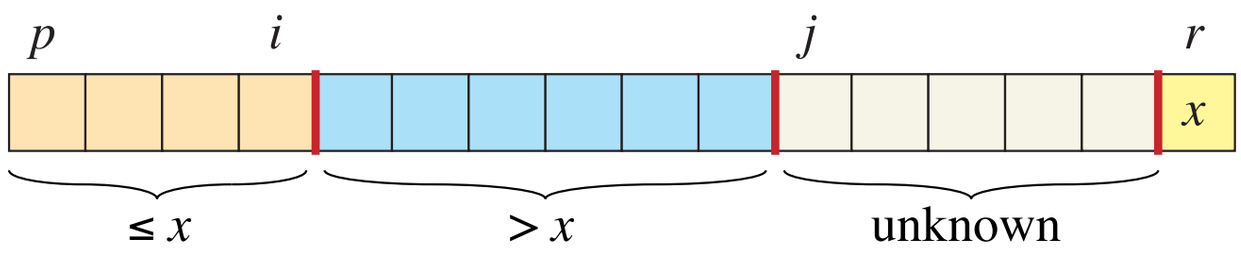
  - If $i + 1 \le k \le j - 1$ then $A[k] > x$
  - If $k = r$ then $A[k] = x$



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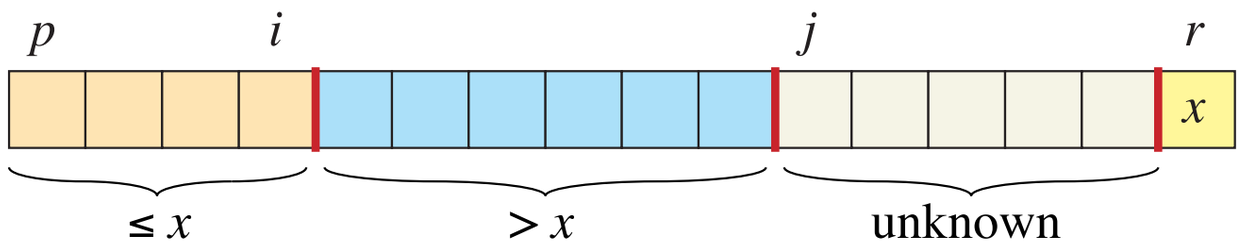
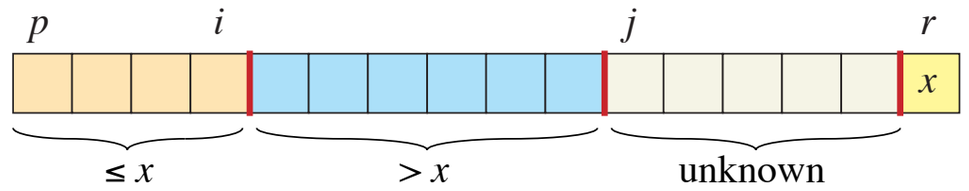
This should hold for any array index k, at the beginning of each for loop iteration.



PARTITION loop invariant

- Let us check the loop invariant holds.

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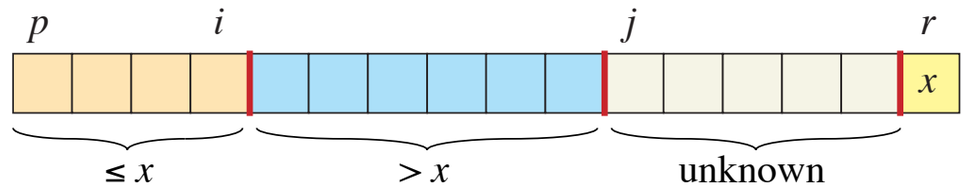


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Initialization:

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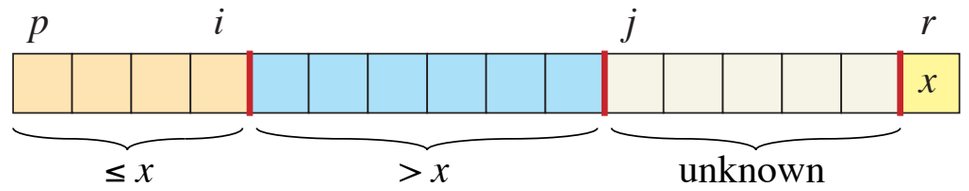


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Maintenance:

PARTITION loop invariant

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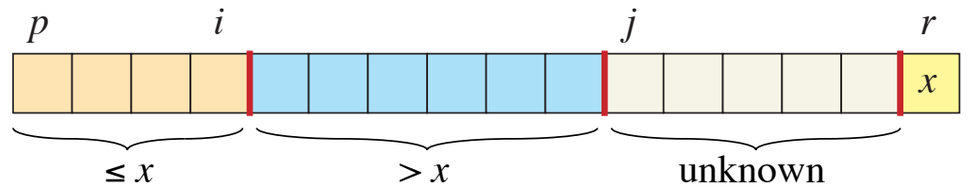


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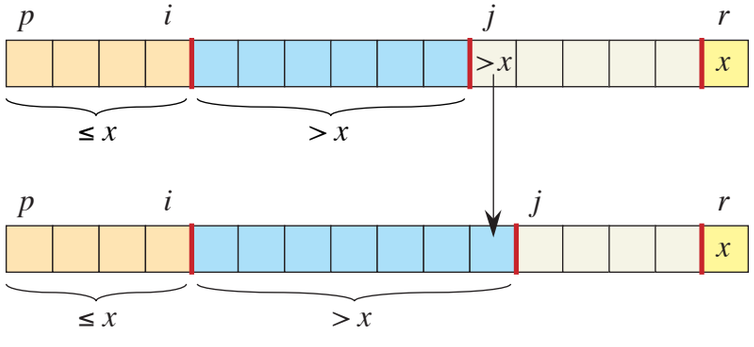
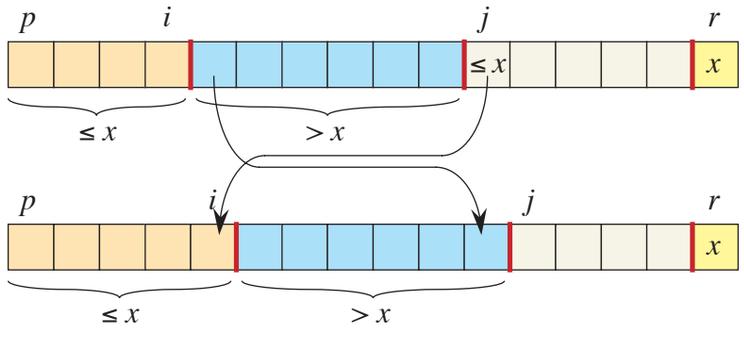
Maintenance:

Case 1



Case 2



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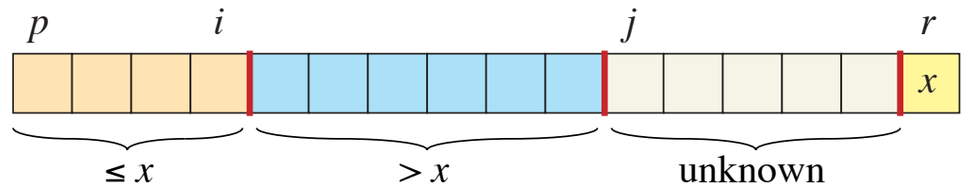


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Termination:

Exercise

- What is the running time for the Partition operation?

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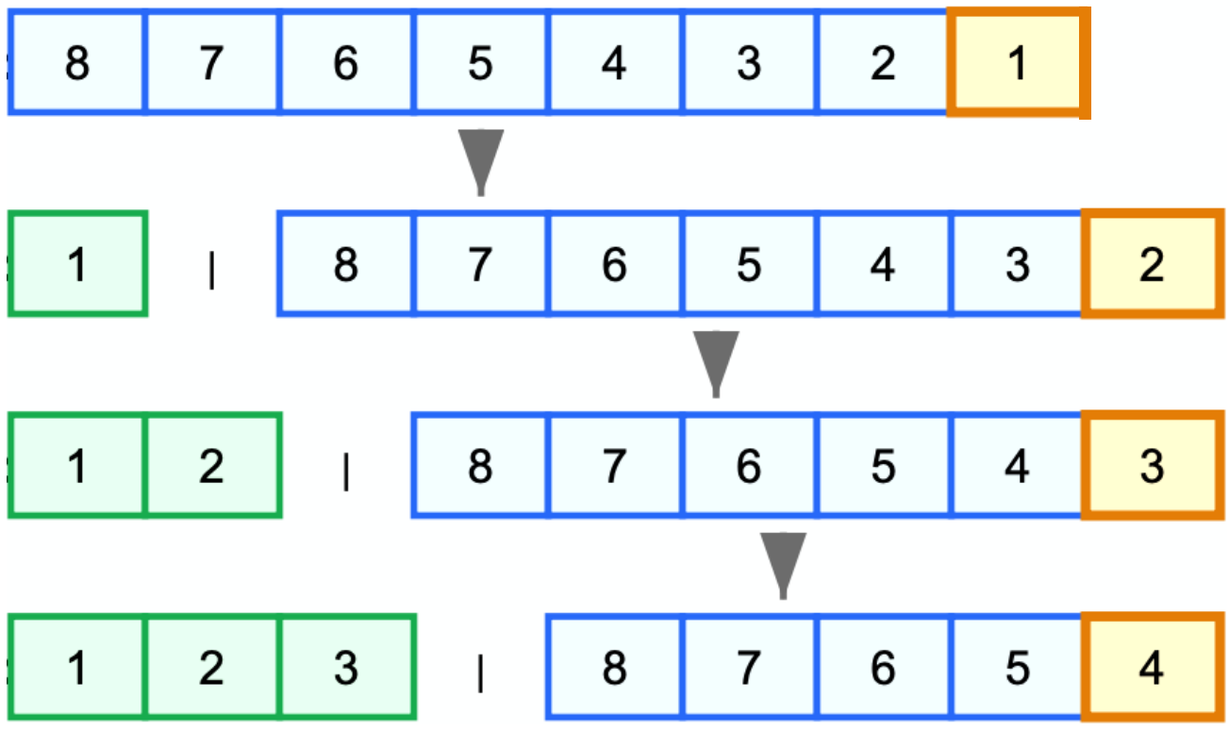
$\Theta(n)$ for an array of size n.

Since we run through the array, making one comparison check per element.

Quick sort performance

- The running time of Quick sort will depend on how balanced each partitioning is.
    - **Worst case partitioning:**

      One subarray with *n-1* elements, one with *0* and the pivot.



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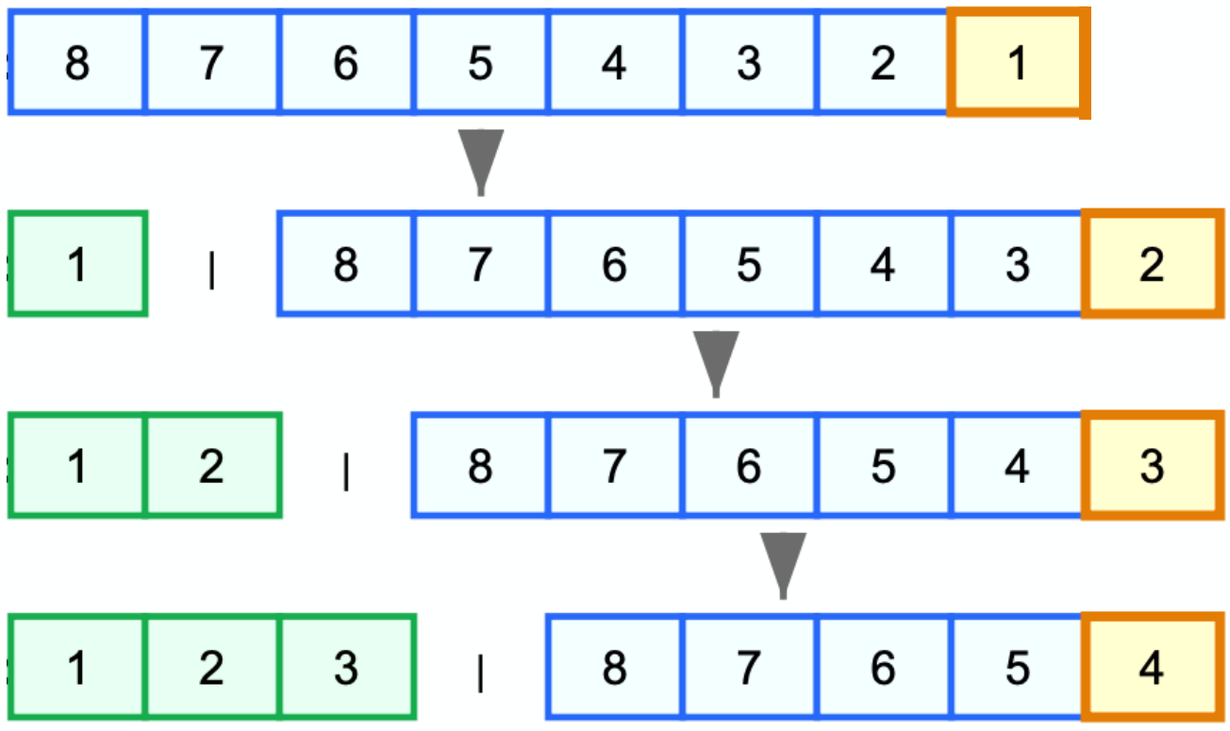

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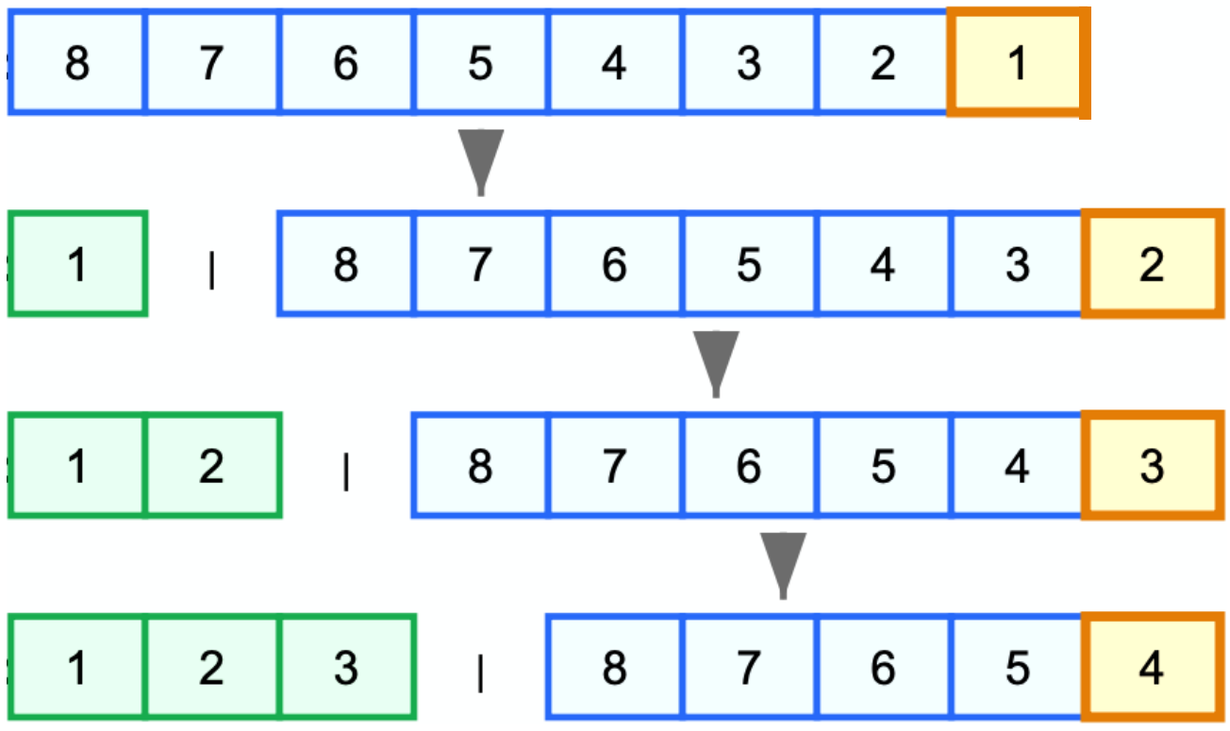

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Similar to INSERTION SORT.

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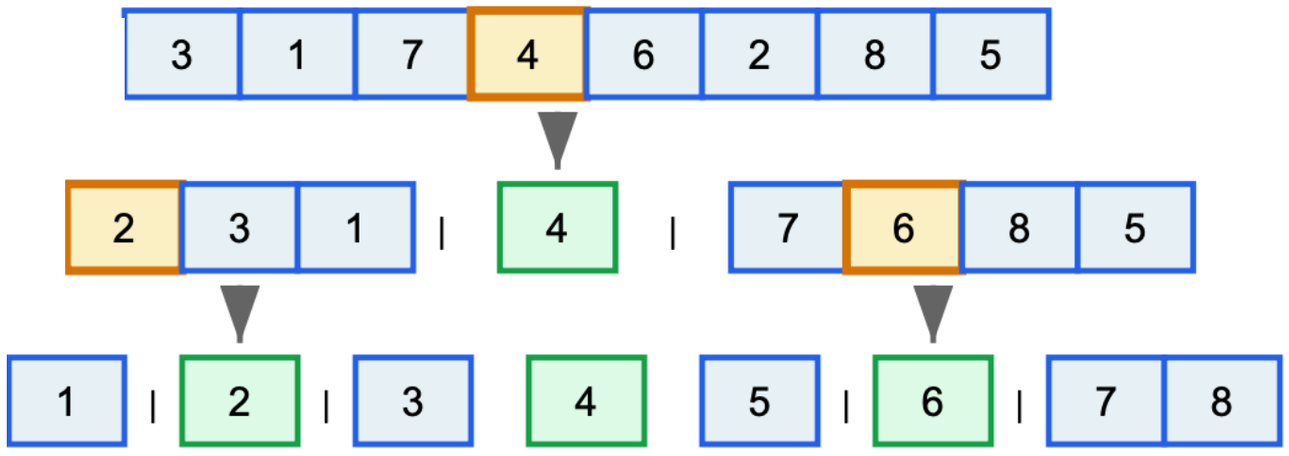
      Each subarray split in half at each iteration.



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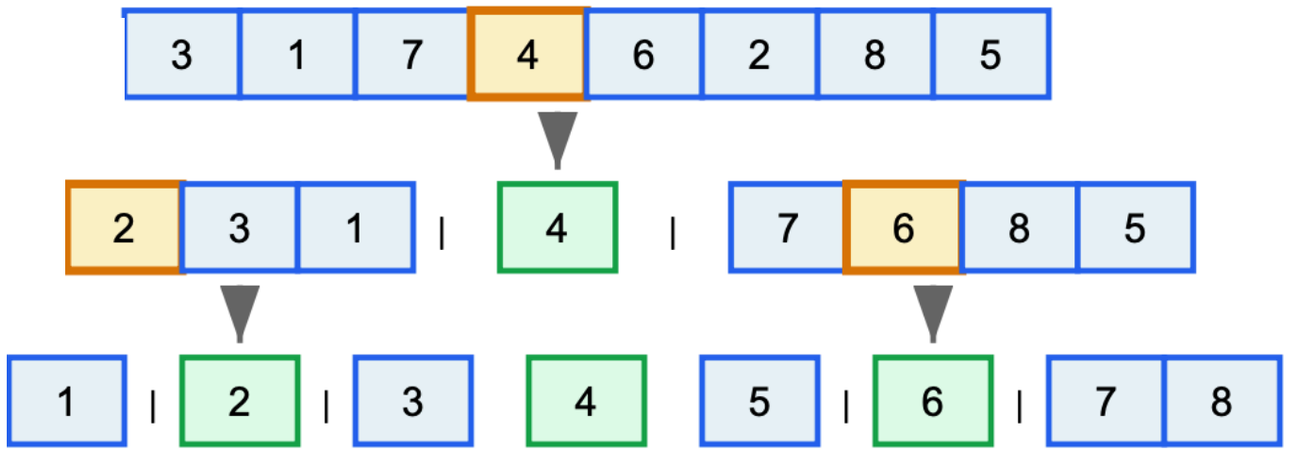

$$T(n) = 2T(n/2) + T(0) + \Theta(n)$$

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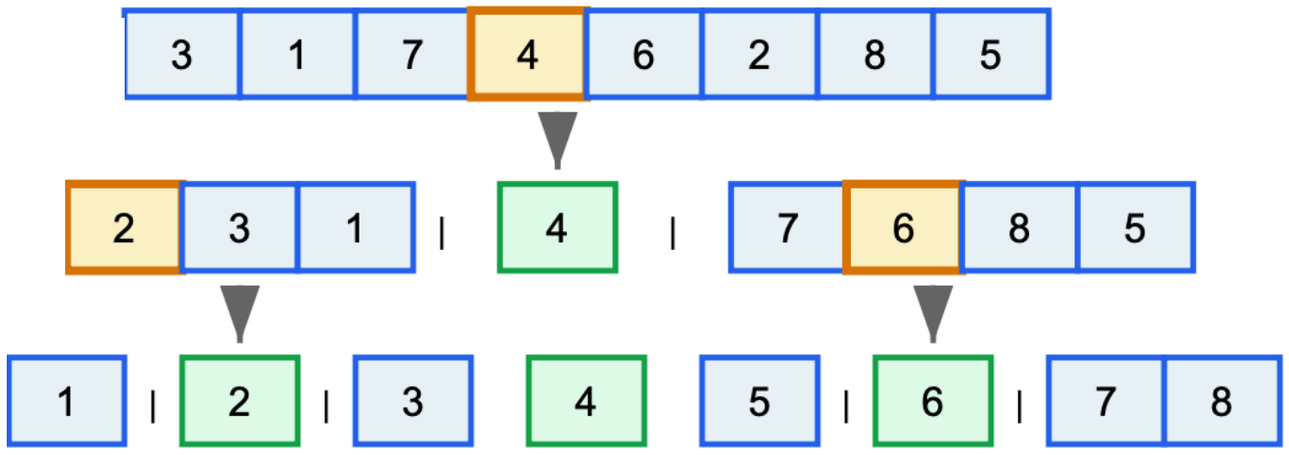

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Similar to Merge Sort

Homework (due with Assignment 1)

- Banks record transactions with respect to the time of the transaction, but customers like to receive their bank statements with checks listed in order by check number. As a checkbook is ordered by check number, people tend to use their checks in order by check number. Furthermore, merchants usually cash checks reasonably promptly.

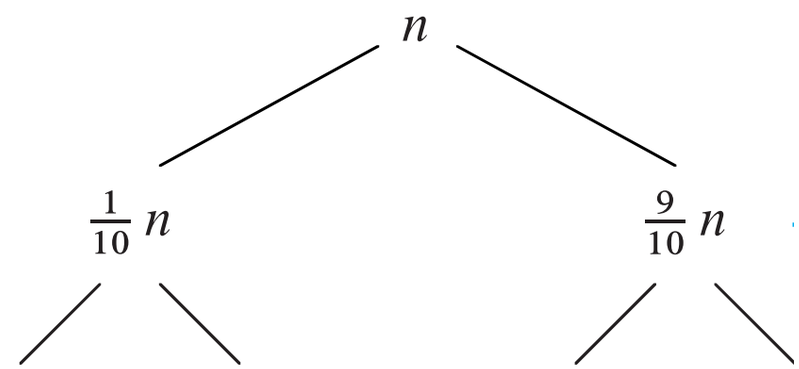
  Thus, the problem of converting time-of-transaction ordering to check-number ordering corresponds to the problem of sorting almost-sorted input.

  Explain persuasively why the INSERTION SORT algorithm might tend to beat QUICK SORT on this problem.

QUICK SORT average performance

- Let us compute the running time for a **9 to 1 partition**

$$T(n) = T(9n/10) + T(n/10) + \Theta(n)$$



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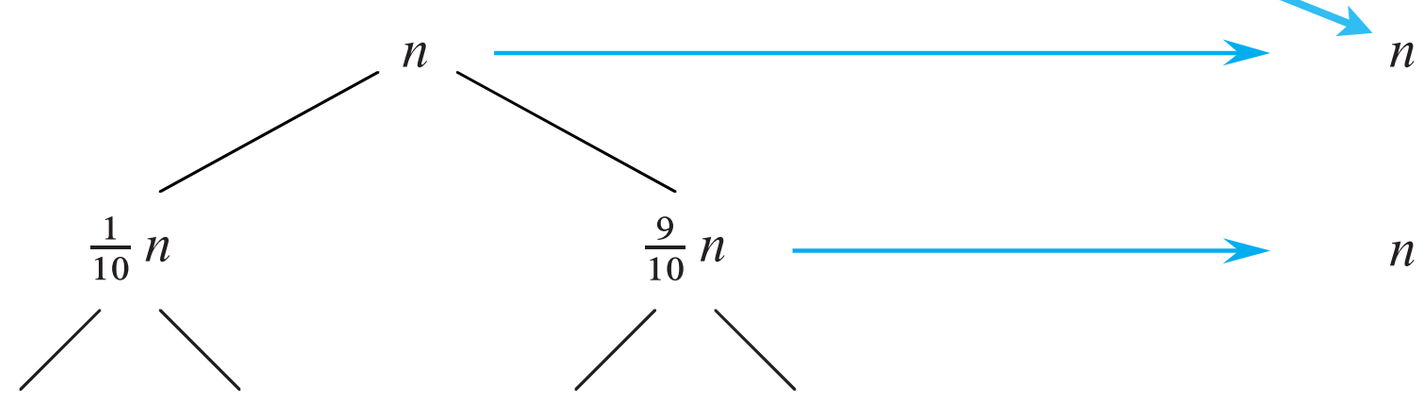

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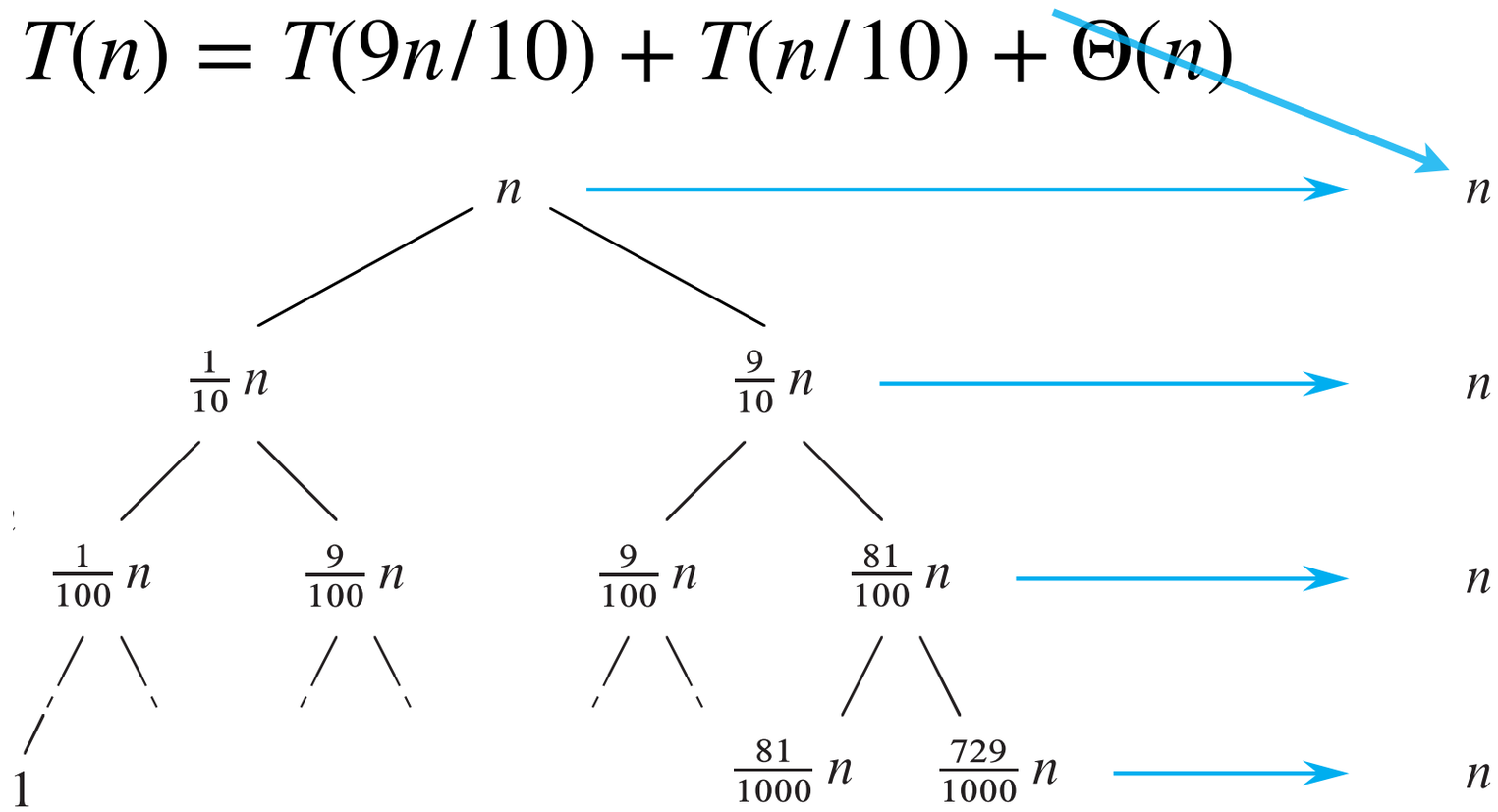

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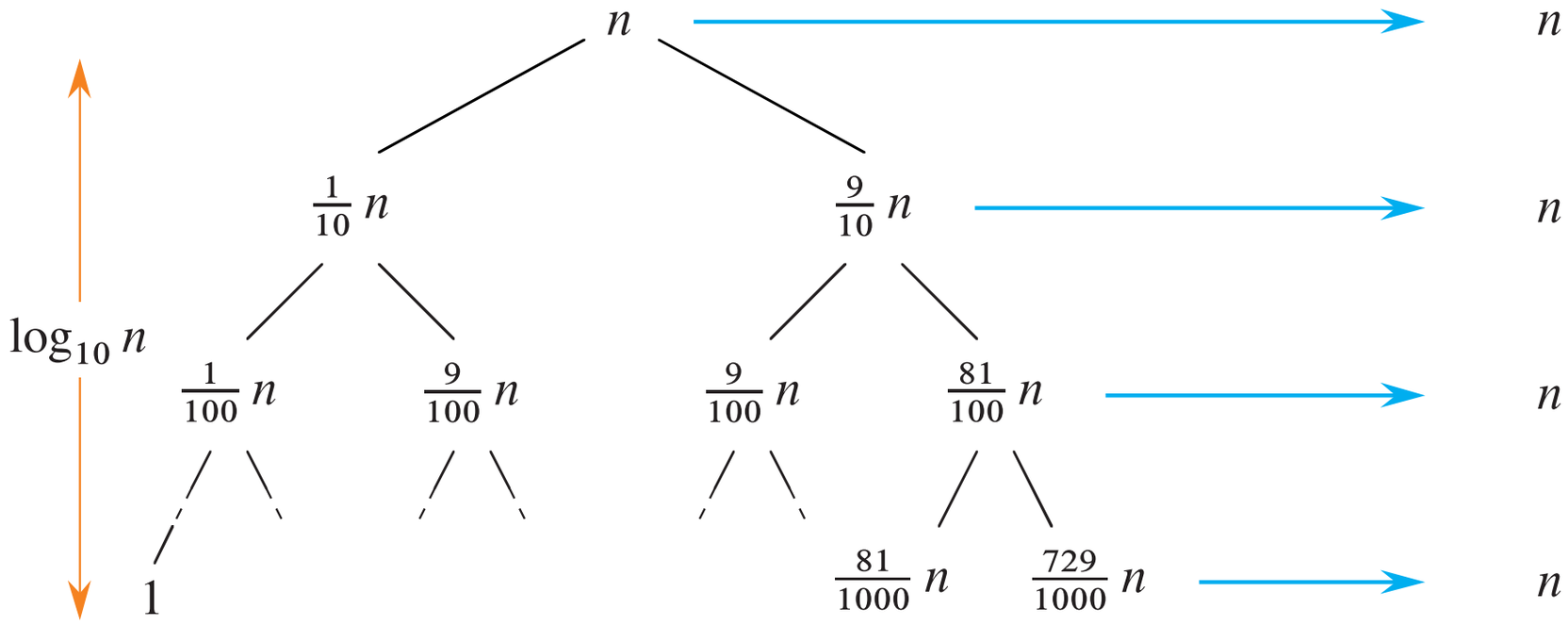

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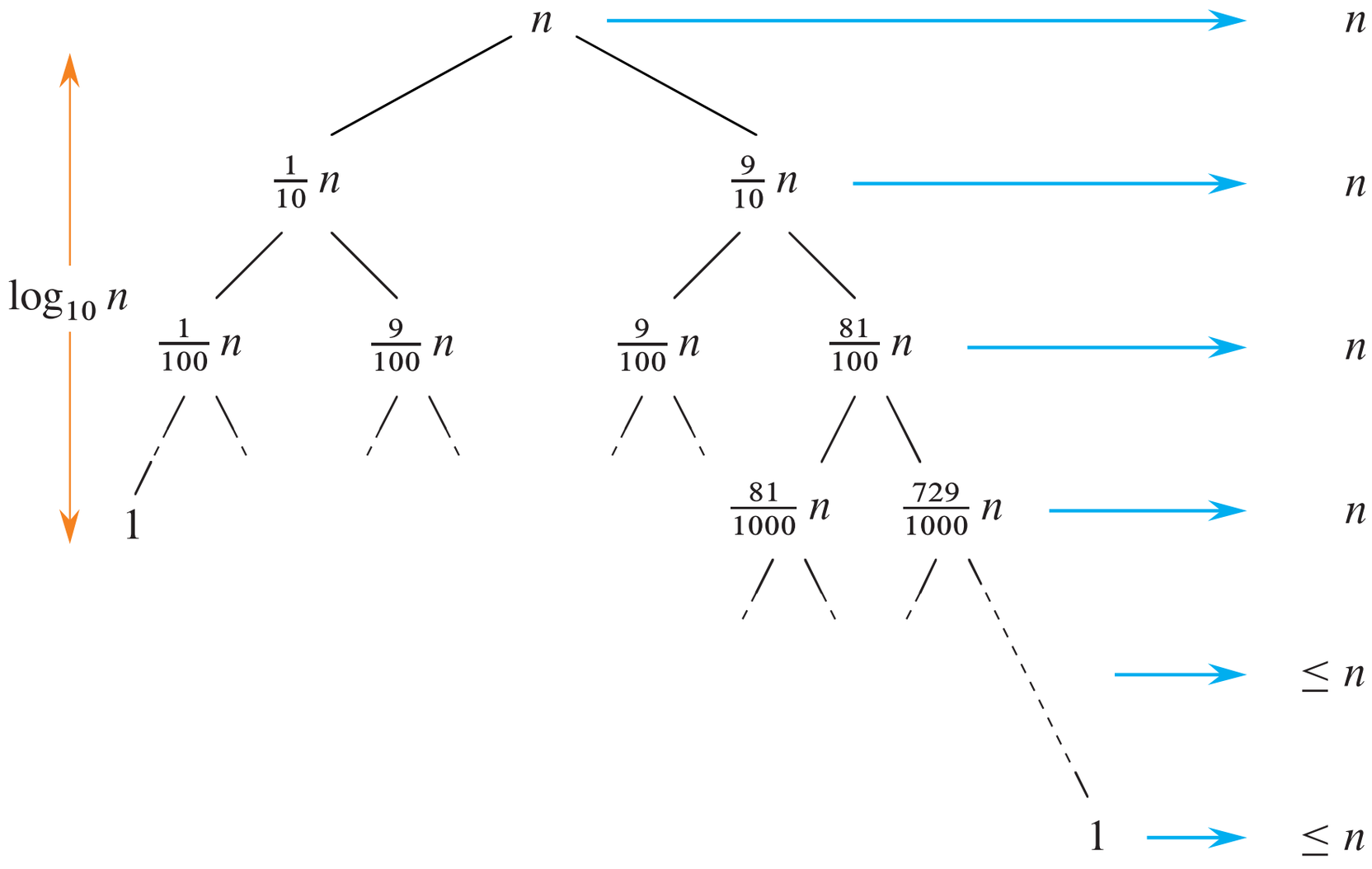

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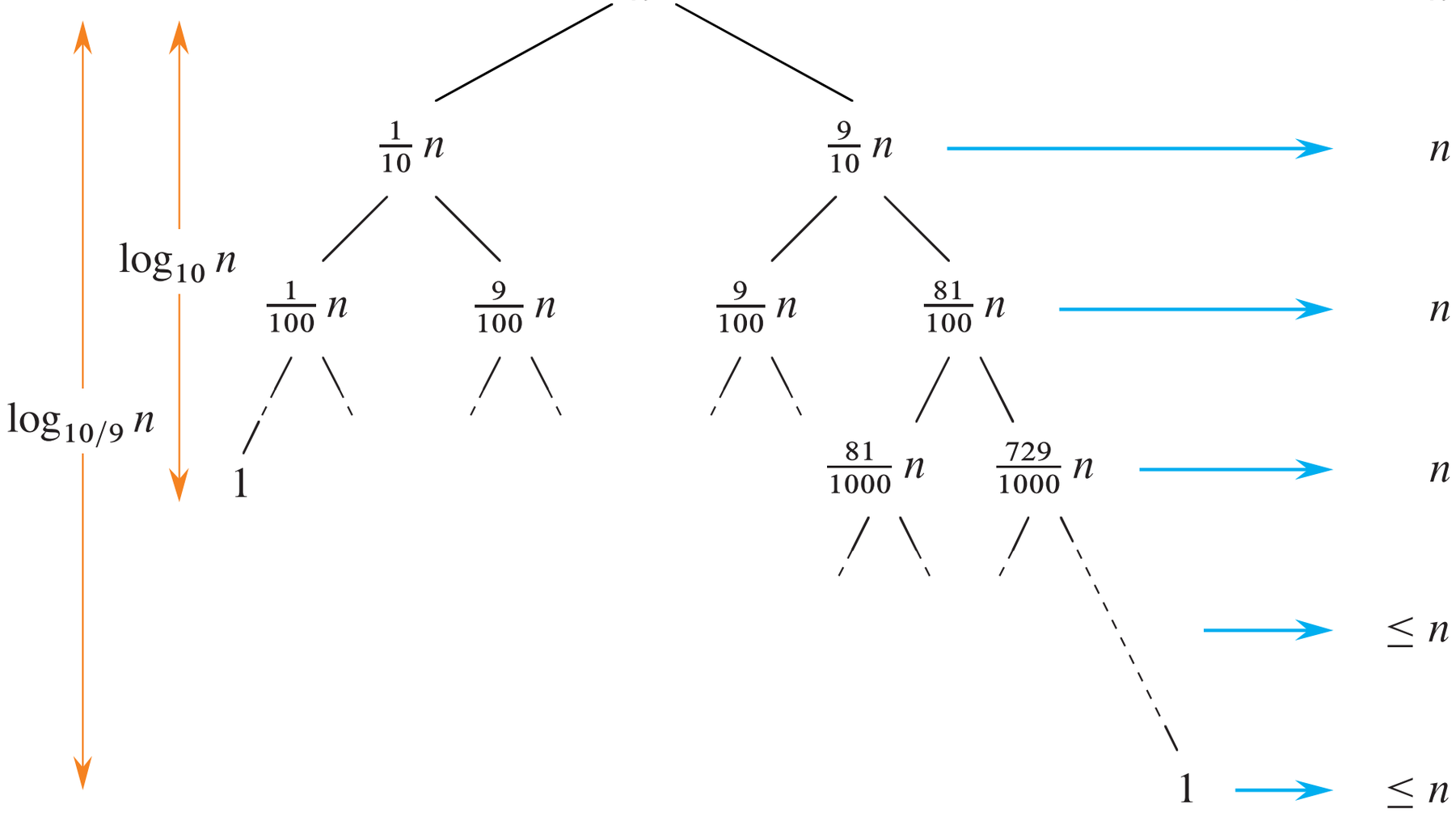

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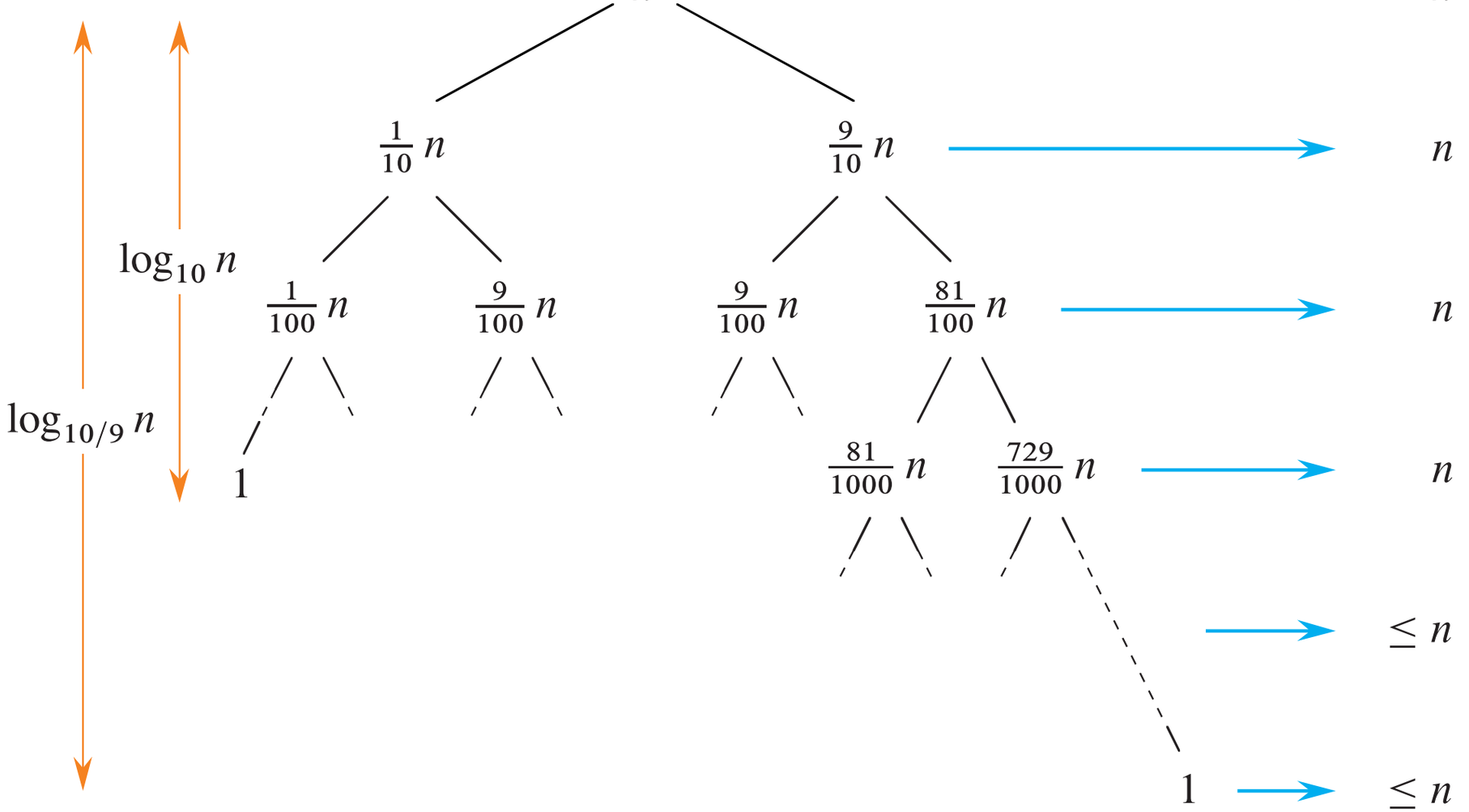
$$T(n) = T(9n/10) + T(n/10) + \Theta(n) = \Theta(nlog_{10/9}(n))$$



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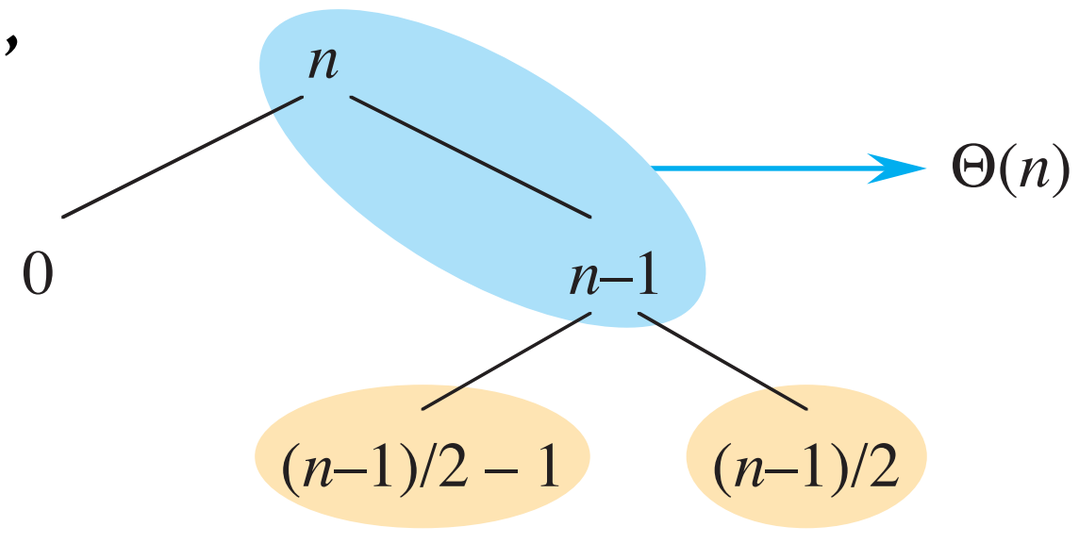

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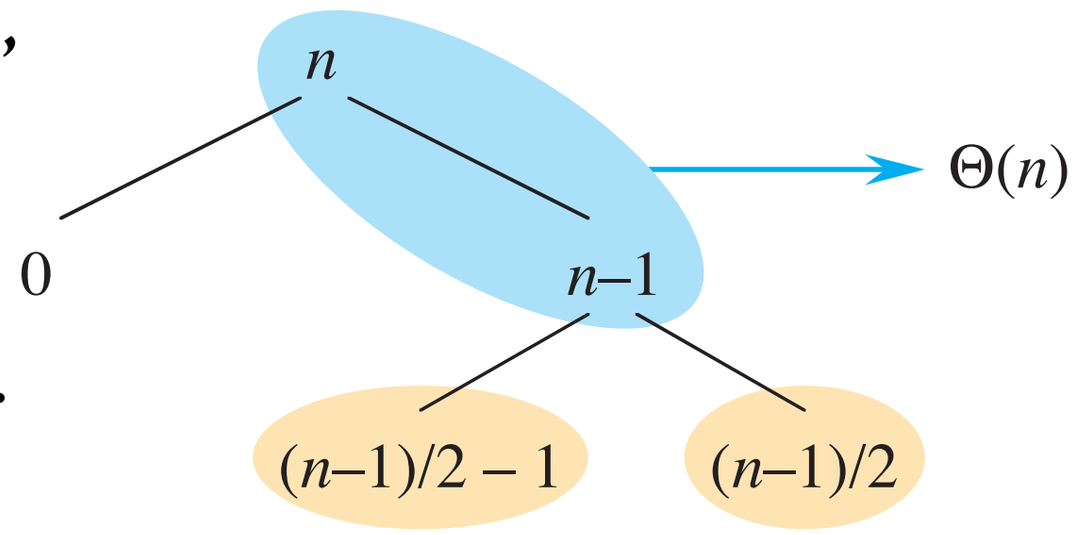

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- Doing two such splits, is similar to a single split, in the best case.

We conclude doing one good and one bad split, is about twice as slow as doing only good splits.

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$$T(n) = \alpha \cdot O(nlogn) = O(nlogn)$$

- Assuming a random distribution of entries, the average running time of QUICKSORT is $O(nlogn)$.

Homework (due with Assignment 1)

- What is the running time of QUICK SORT when all elements of the array have the same value?

Randomized QUICK SORT

- Since QUICKSORT tend to work best with randomized entries, we might consider randomizing the entries first.

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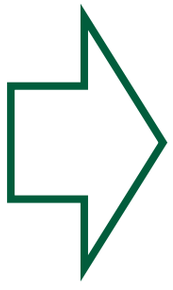

```
RANDOMIZED-PARTITION(A, p, r)
1  i = RANDOM(p, r)
2  exchange A[r] with A[i]
3  return PARTITION(A, p, r)
```

Exercise

- Compare the two partition functions:

```
PARTITION(A, p, r)
1   x = A[r]
2   i = p − 1
3   for j = p to r − 1
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8   return i + 1
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```
HOARE-PARTITION(A, p, r)
 1   x = A[p]
 2   i = p − 1
 3   j = r + 1
 4   while TRUE
 5       repeat
 6           j = j − 1
 7       until A[j] ≤ x
 8       repeat
 9           i = i + 1
10       until A[i] ≥ x
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We swap every element smaller than the pivot with the partition boundary.

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13      else return j
```

Makes less swaps: ensures both elements moved are on the wrong side.

Exercise

- You are given n red and n blue water jugs, all of different shapes and sizes. All the jugs hold different amounts of water, and you cannot tell from the size of a jug how much water it holds. But, for every jug of one color, there is a jug of the other color that holds the same amount of water.
- Your task is to group the jugs into pairs of red and blue jugs that hold the same amount of water.
- Your goal is to find an algorithm that makes a minimum number of pairwise comparisons between different colored jugs (you are not allowed to compare two red jugs or two blue jugs) to determine the grouping.

Exercise

- **Divide:**
  - Pick any red jug R as a pivot
  - Compare R with each blue jug to partition the blue jugs into three groups:
    - Blue jugs smaller than R
    - The blue jug B that equals R (there's exactly one)
    - Blue jugs larger than R
  - Now use the matching blue jug B to partition the remaining red jugs:
    - Red jugs smaller than B
    - Red jugs larger than B
- **Conquer:**
  - Recursively solve the subproblem of smaller and larger jugs
  - The pivot pair (R, B) is already matched

Bucket Sort

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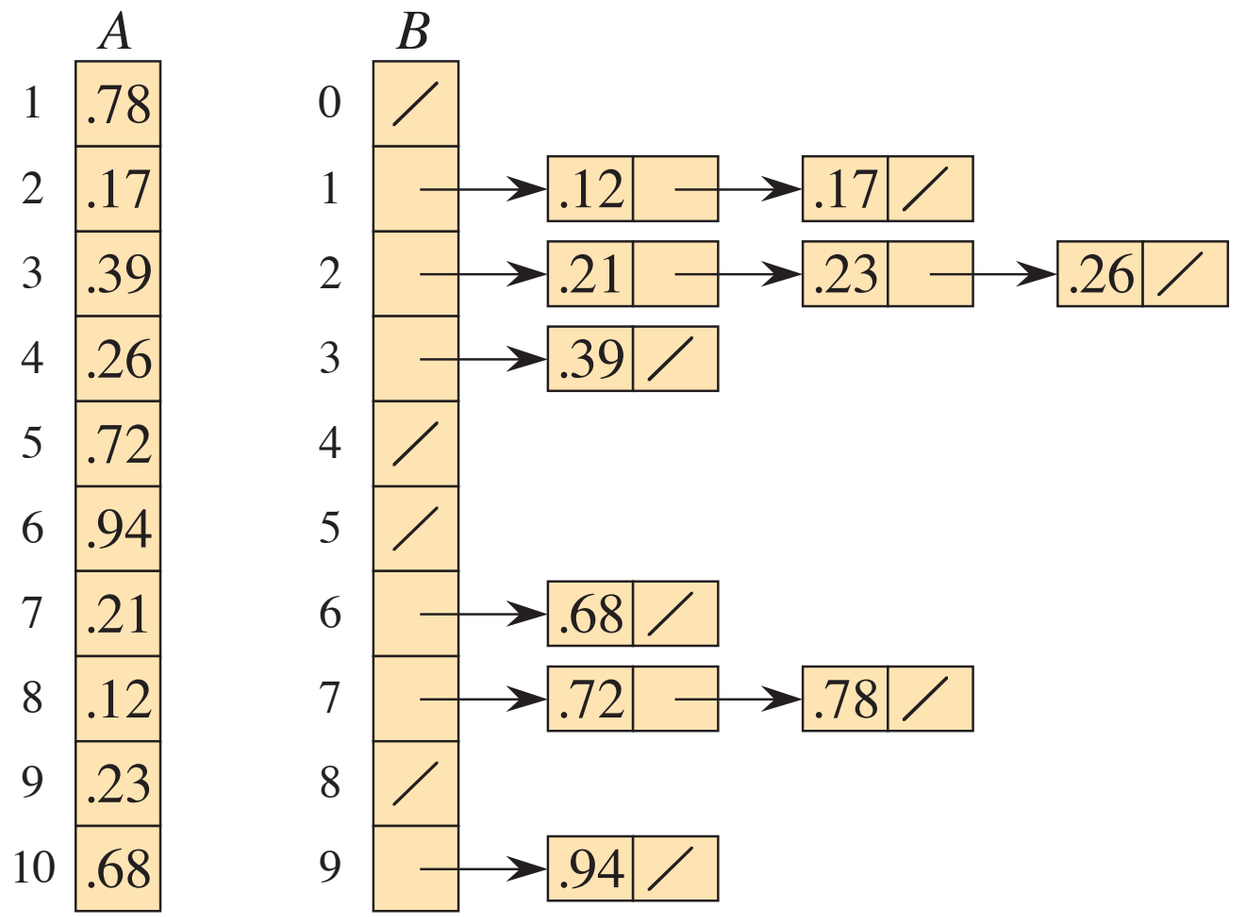

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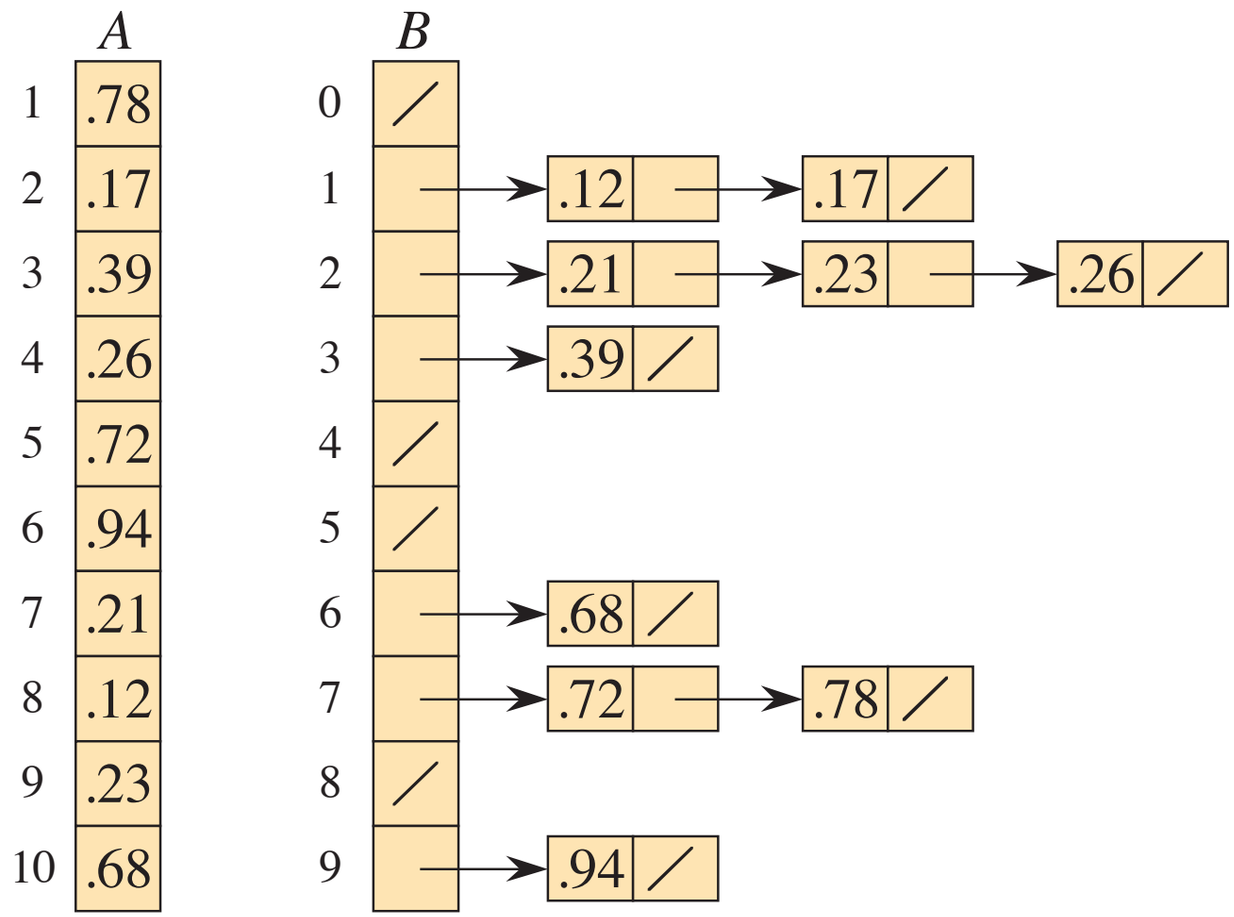
Reminiscent of Hash tables :)



BUCKET SORT

```
BUCKET-SORT(A, n)
1  let B[0 : n − 1] be a new array
2  for i = 0 to n − 1
3      make B[i] an empty list
4  for i = 1 to n
5      insert A[i] into list B[⌊n · A[i]⌋]
6  for i = 0 to n − 1
7      sort list B[i] with insertion sort
8  concatenate the lists B[0], B[1], . . . , B[n − 1] together in order
9  return the concatenated lists
```

Homework (due with Assignment 1)

- What is the running time of QUICK SORT when all elements of the array have the same value?
- An array A of size is $n>10$ is filled as follows:

  For each element $A[i]$, choose two random variables x_i and y_i uniformly and independently from $[0,1)$. Then set:

$$A[i] = \frac{\lfloor 10x_i \rfloor}{10} + \frac{y_i}{n}$$

  Modify BUCKET SORT so that it sorts the array A in $O(n)$ expected time.